Lab 2

DHCP Server and Windows Deployment Services (WDS)

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Installing and configuring a DHCP server, and deploying Windows 10 over the network using PXE

1. Overview and Objectives

In this lab a DHCP server is installed to assign IP addresses automatically across the network. In the second part, Windows Deployment Services (WDS) is installed so that Windows operating systems can be deployed over the network. This procedure is common in large organizations and institutions, such as schools and medium-to-large businesses that need to deploy dozens of computers quickly without installing each one from a USB drive.

After completing this lab, the following objectives are achieved:

- Install the DHCP Server role on Windows Server 2022.
- Create, configure and activate a DHCP scope (address range, exclusion range, lease duration and DNS options).
- Install and configure the Windows Deployment Services role.
- Add a Windows 10 boot image (boot.wim) and install image (install.wim) to WDS.
- Deploy Windows 10 to a blank virtual machine using PXE boot.

2. Lab Environment and Requirements

Item	Details
Server virtual machine	DC01 (VMware Workstation)
Server operating system	Windows Server 2022 Datacenter Evaluation
Server computer name	WIN-FEVN8KFH14H
Server resources	4 GB RAM, 4 processors, 250 GB disk
Client virtual machine	Win 10 WDS (blank VM, no operating system, CD/DVD left empty)
Network	Both VMs on the same VMware LAN Segment
Installation media	Windows 10 ISO (mounted on DC01 to extract install.wim and boot.wim)

Roles installed	DHCP Server; Windows Deployment Services (Deployment Server and Transport Server)
Network (scope) name	VLAN 1 (network 192.168.1.0/24)

3. Part 1: DHCP Server Installation and Configuration

3.1 Installing the DHCP Server role

Step 1: Server Manager was opened and "Add roles and features" was clicked.

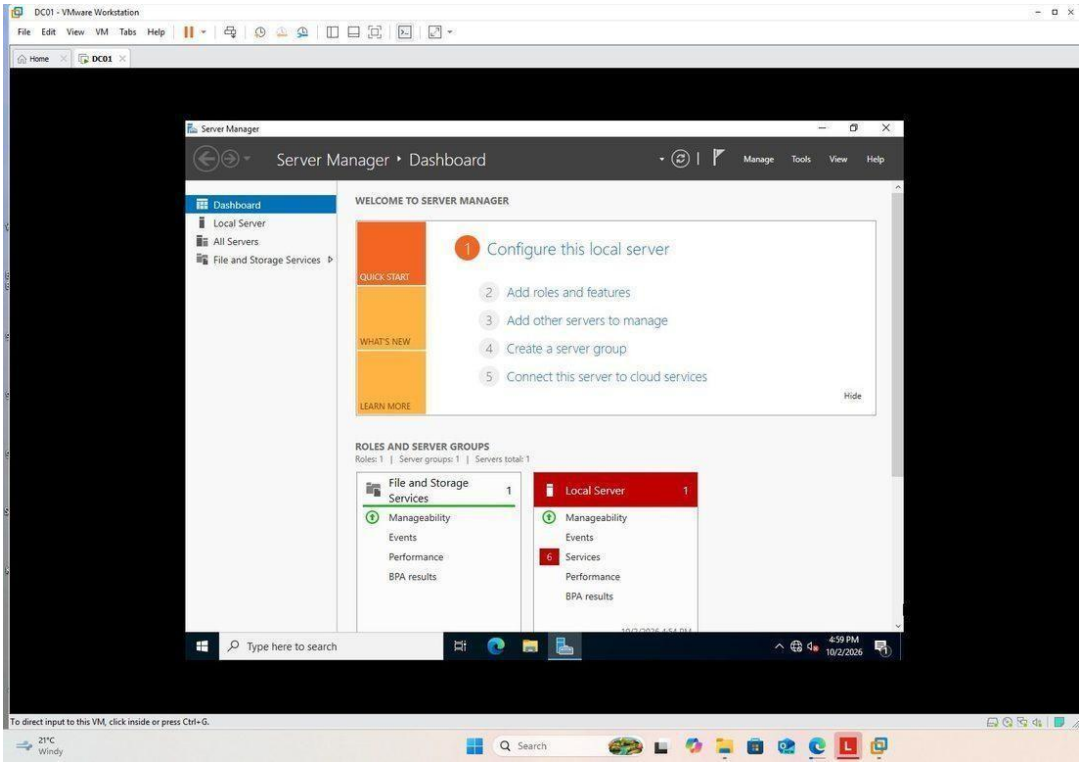


Figure 1: Server Manager dashboard on DC01

Step 2: Next was clicked to proceed through the "Before you begin" page of the wizard, and "Rolebased or feature-based installation" was selected.

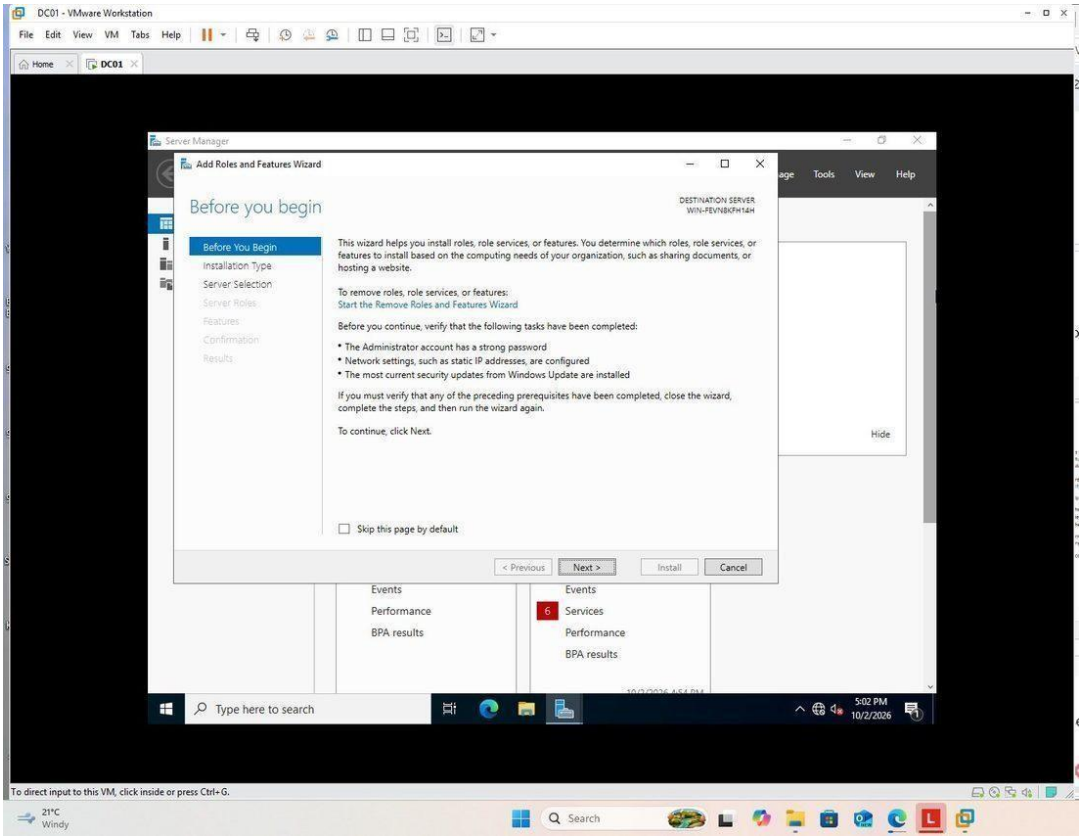


Figure 2: Add Roles and Features Wizard: Before you begin

Step 3: On "Select destination server", "Select a server from the server pool" was chosen and the local server (WIN-FEVN8KFH14H) was selected.

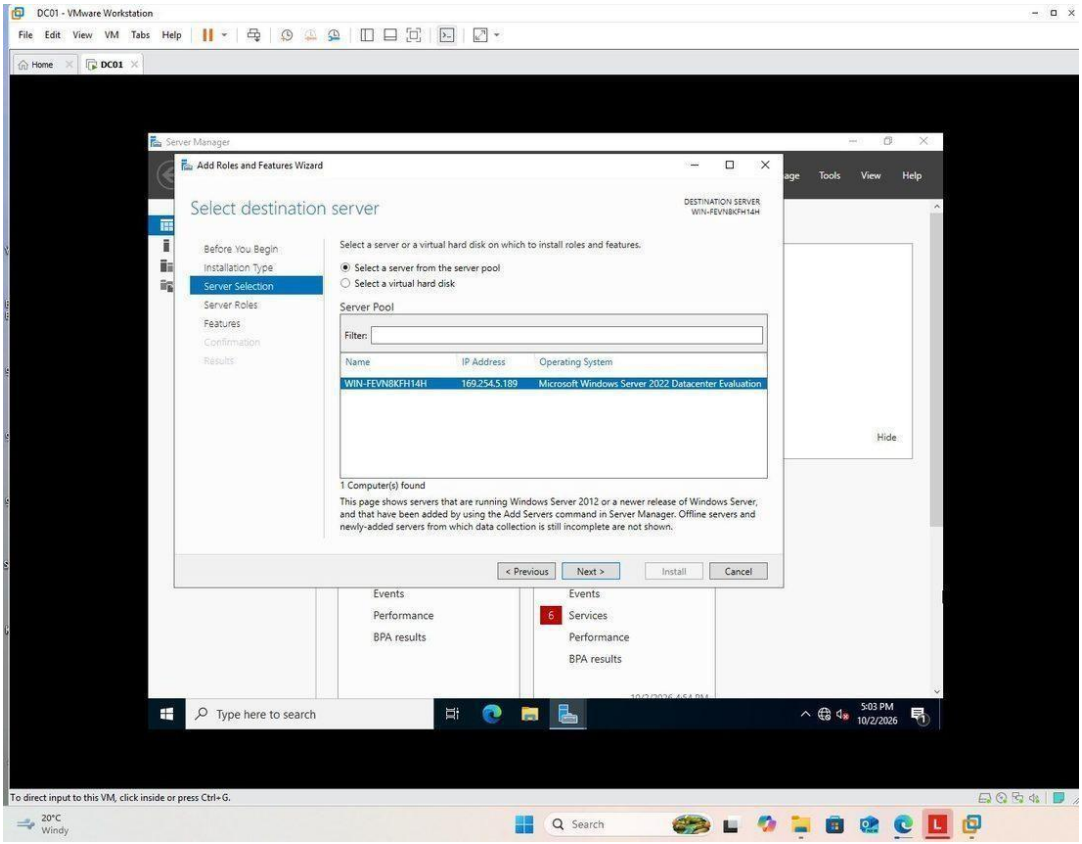


Figure 3: Select destination server

Step 4: On "Select server roles", the DHCP Server check box was ticked and the additional features offered were accepted.

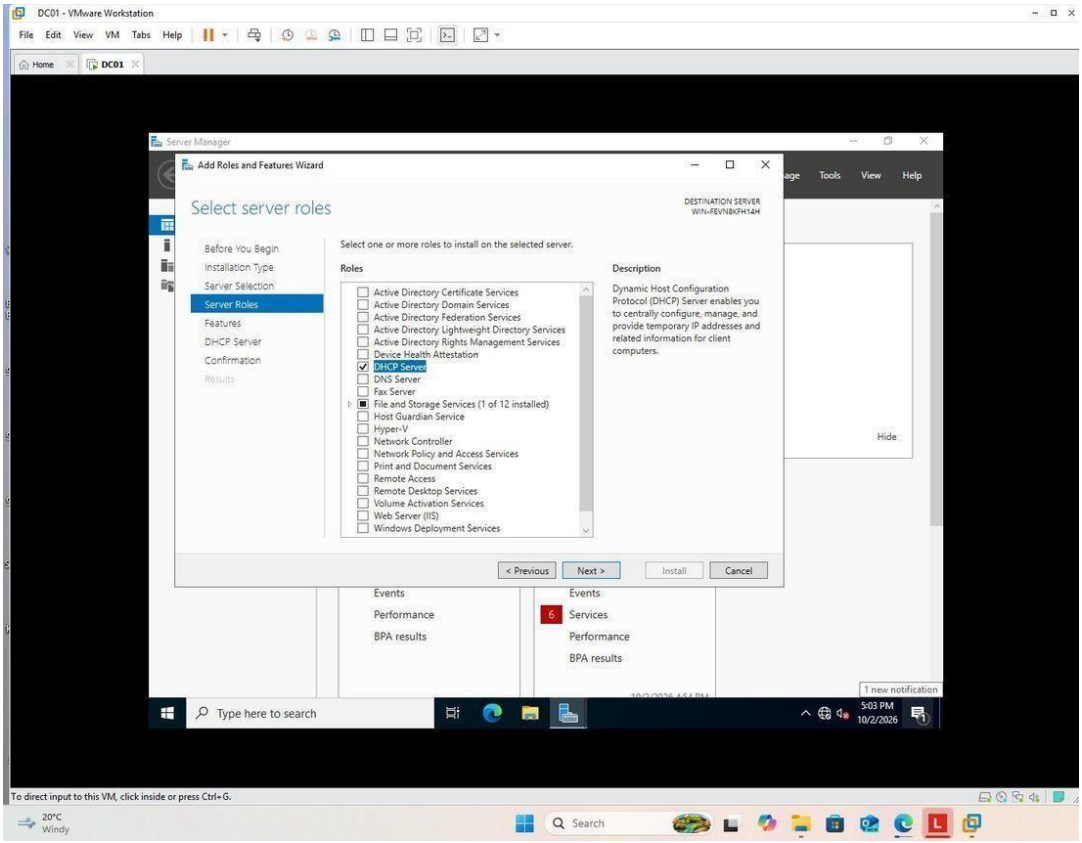


Figure 4: DHCP Server role selected

Step 5: The selections were reviewed on the confirmation page and Install was clicked.

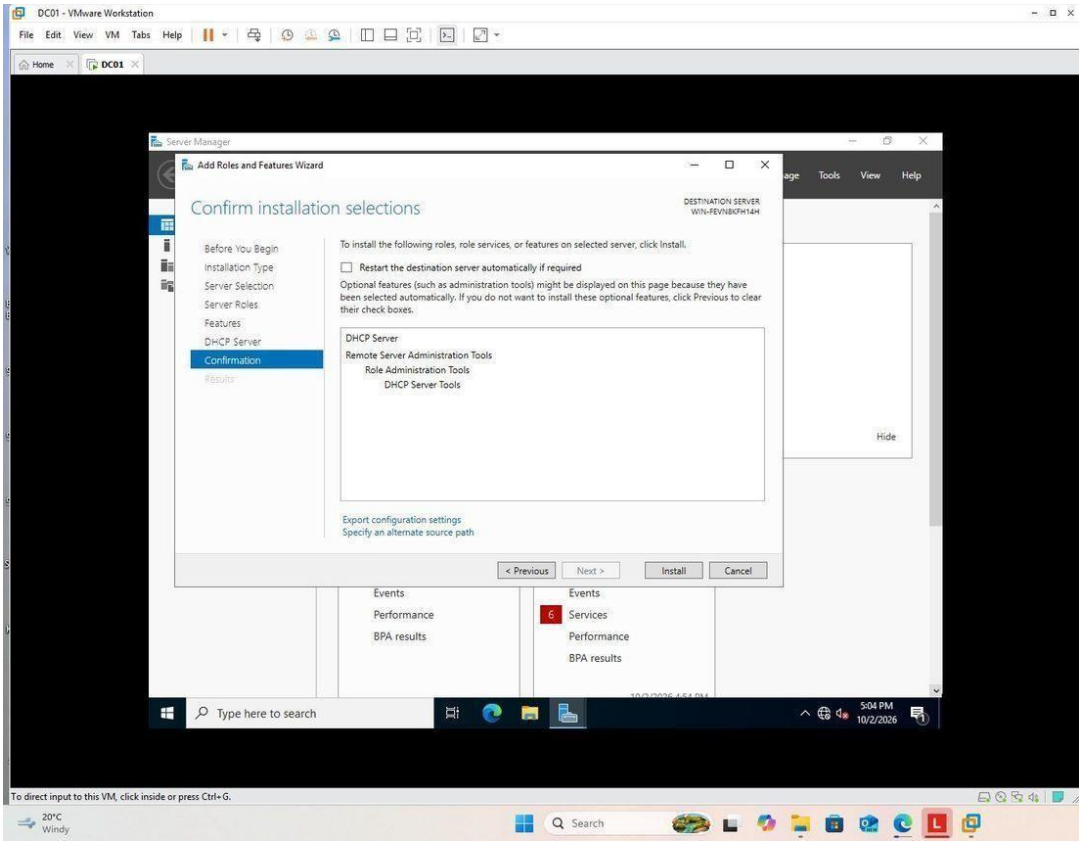


Figure 5: Confirm installation selections (DHCP Server and DHCP Server Tools)

Step 6: When the installation had succeeded, "Complete DHCP configuration" was clicked and the DHCP post-install wizard was launched.

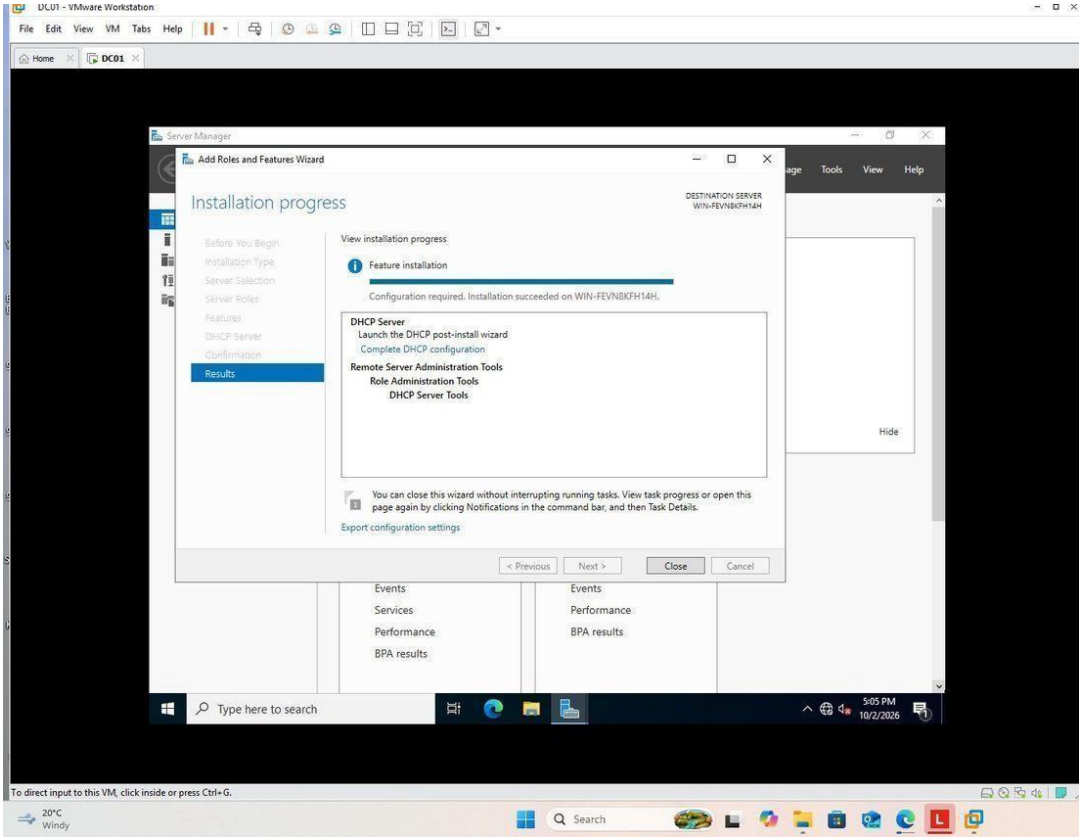


Figure 6: Installation progress: succeeded, with the "Complete DHCP configuration" link

Step 7: Next was clicked in the post-install wizard, then Commit was clicked and the dialog was closed.

> **Important:** A DHCP server must be authorized in Active Directory (AD DS) before it can hand out IP addresses. This is done by the Commit button in the post-install wizard. Without authorization, the DHCP labs cannot be completed.

3.2 Creating and configuring the DHCP scope

Step 8: In Server Manager, Tools was opened and DHCP was selected.

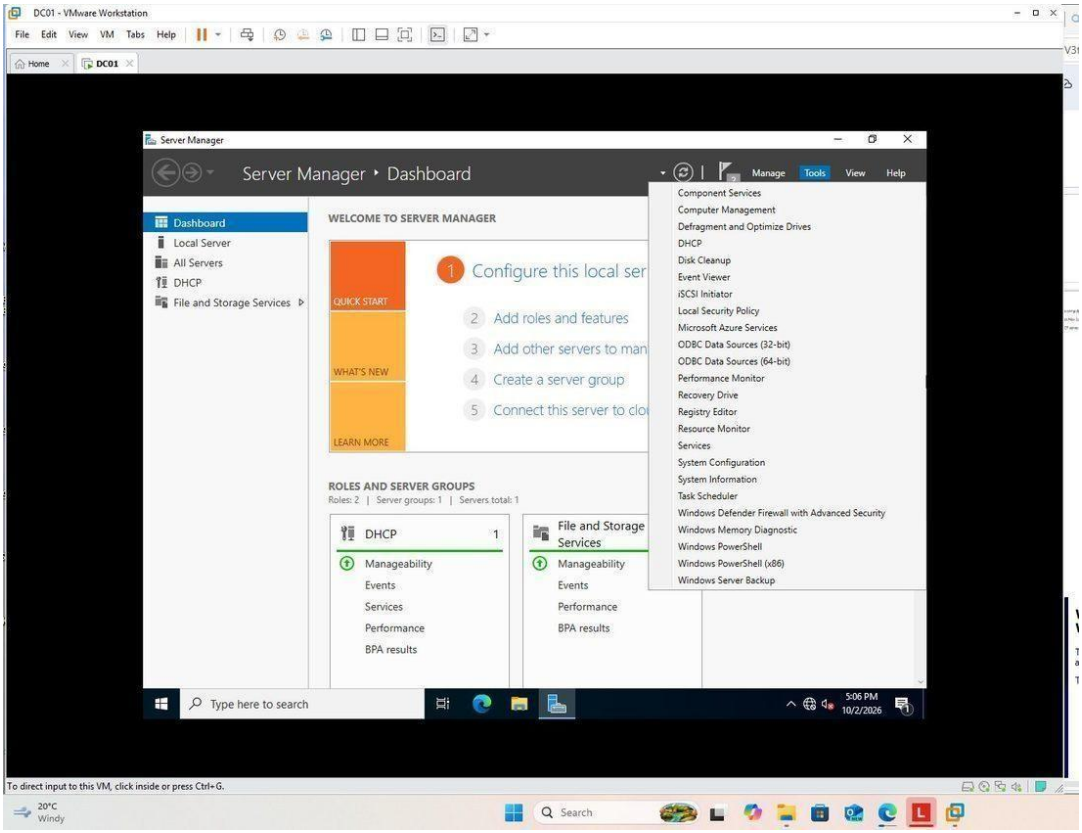


Figure 7: Server Manager Tools menu with DHCP

Step 9: In the DHCP console, IPv4 was right-clicked on the server and "New Scope..." was selected.

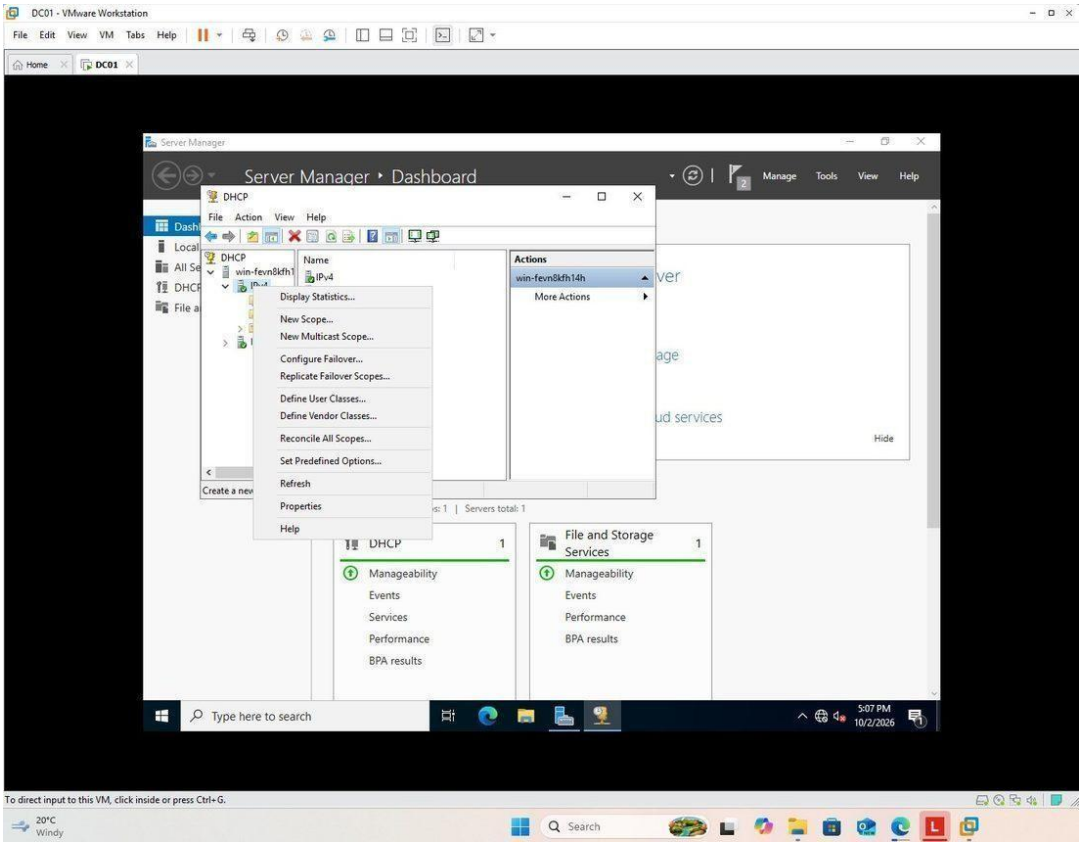


Figure 8: Right-click IPv4 and choose New Scope

Step 10: Next was clicked on the Welcome page of the New Scope Wizard.

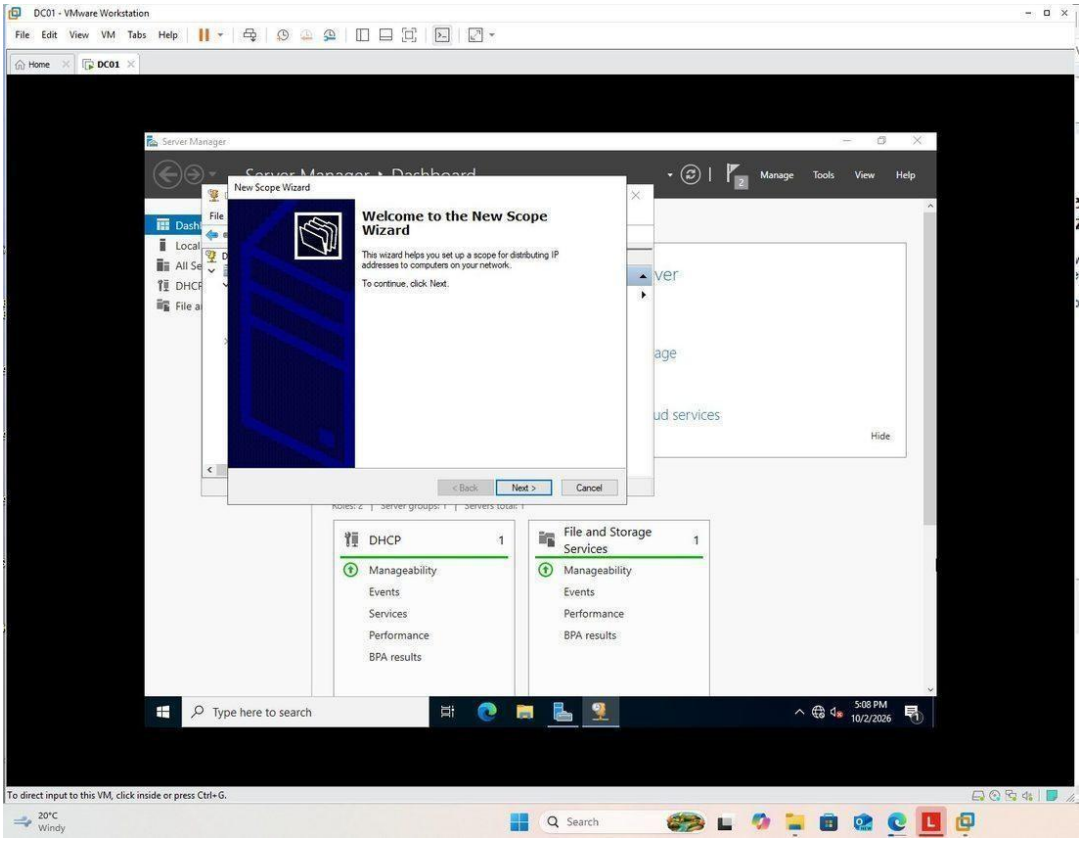


Figure 9: Welcome to the New Scope Wizard

Step 11: The scope was named appropriately. In an organization with many subnets, scopes should be named after the network structure. Here the network is 192.168.1.0/24, which corresponds to VLAN 1 (floor 1, network 1), so the scope was named "VLAN 1".

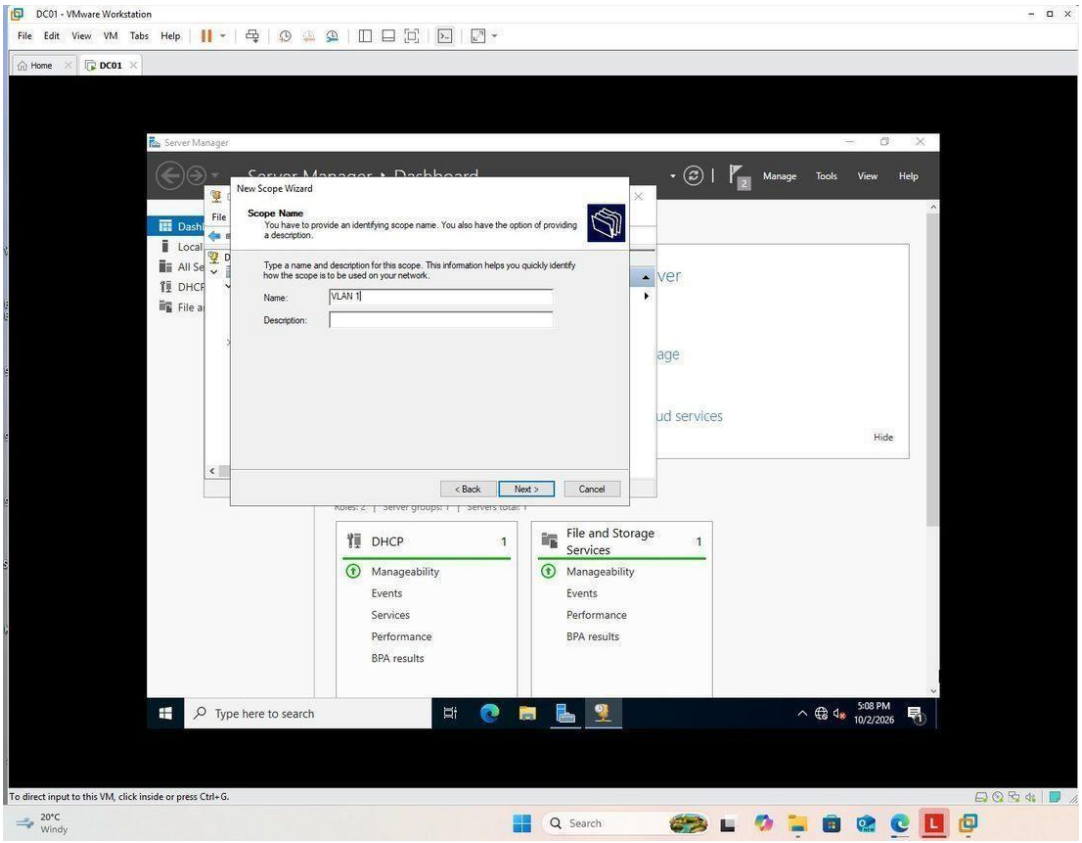


Figure 10: Scope name: VLAN 1

Step 12: The range of IP addresses was allocated by entering the start address 192.168.1.1 and the end address 192.168.1.254. The length was set to 24 and the subnet mask was set to 255.255.255.0.

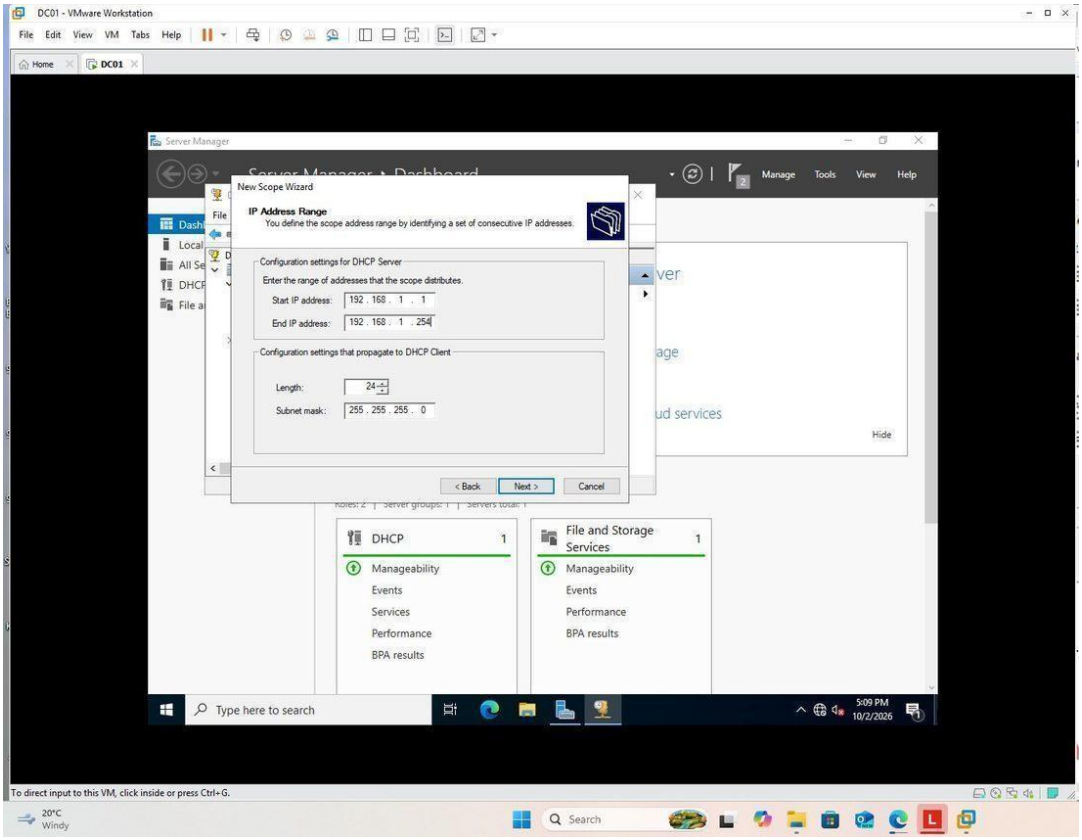


Figure 11: IP address range 192.168.1.1 to 192.168.1.254

Step 13: An exclusion range was created. The first 10 addresses (192.168.1.1 to 192.168.1.10) are normally reserved for gateways (routers) and other fixed devices, so they were excluded from distribution.

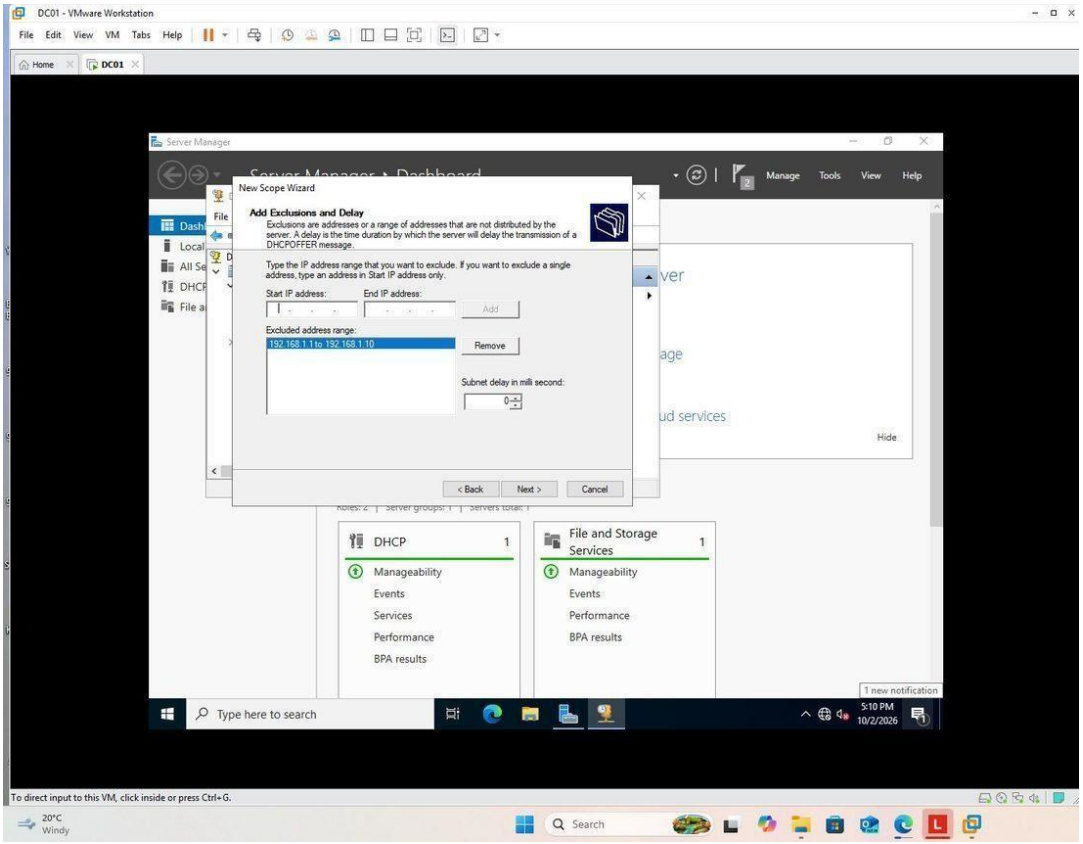


Figure 12: Excluded address range 192.168.1.1 to 192.168.1.10

Step 14: The lease duration was set to 1 hour. IP addresses are leased to clients by DHCP servers for a specified period, after which the leases must be renewed.

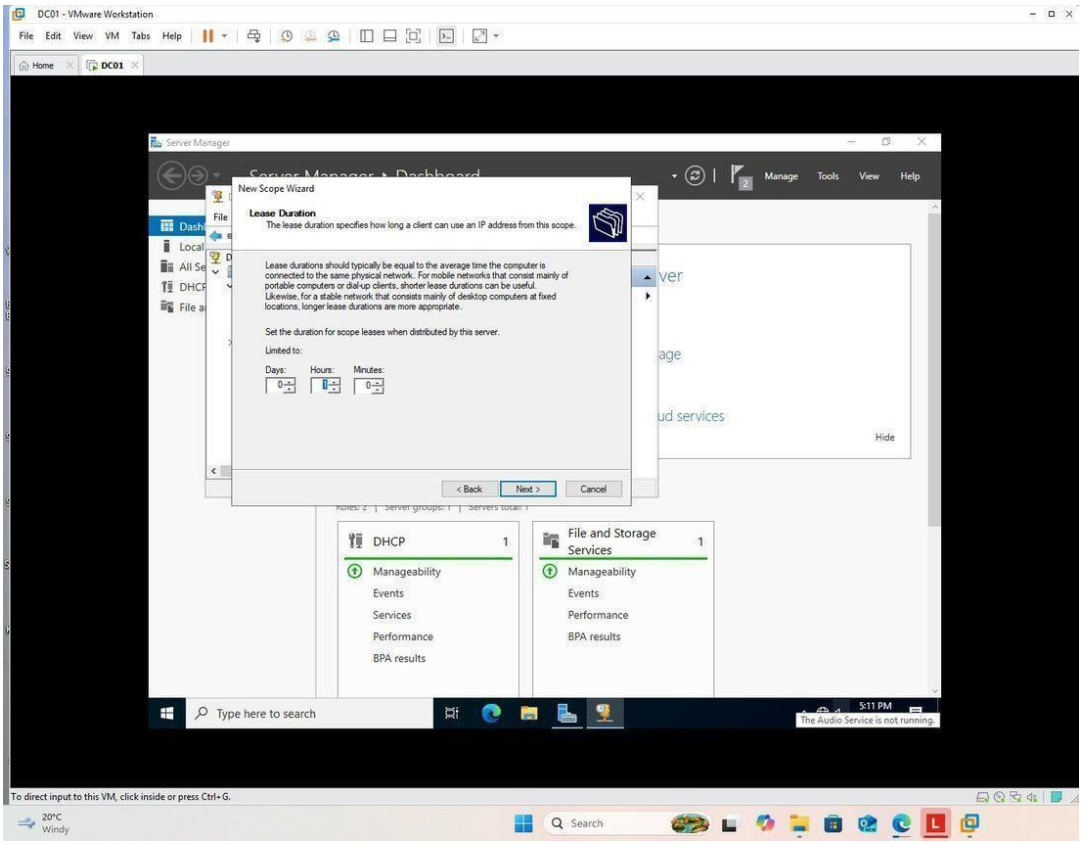


Figure 13: Lease duration: 0 days, 1 hour, 0 minutes

Step 15: "Yes, I want to configure these options now" was selected.

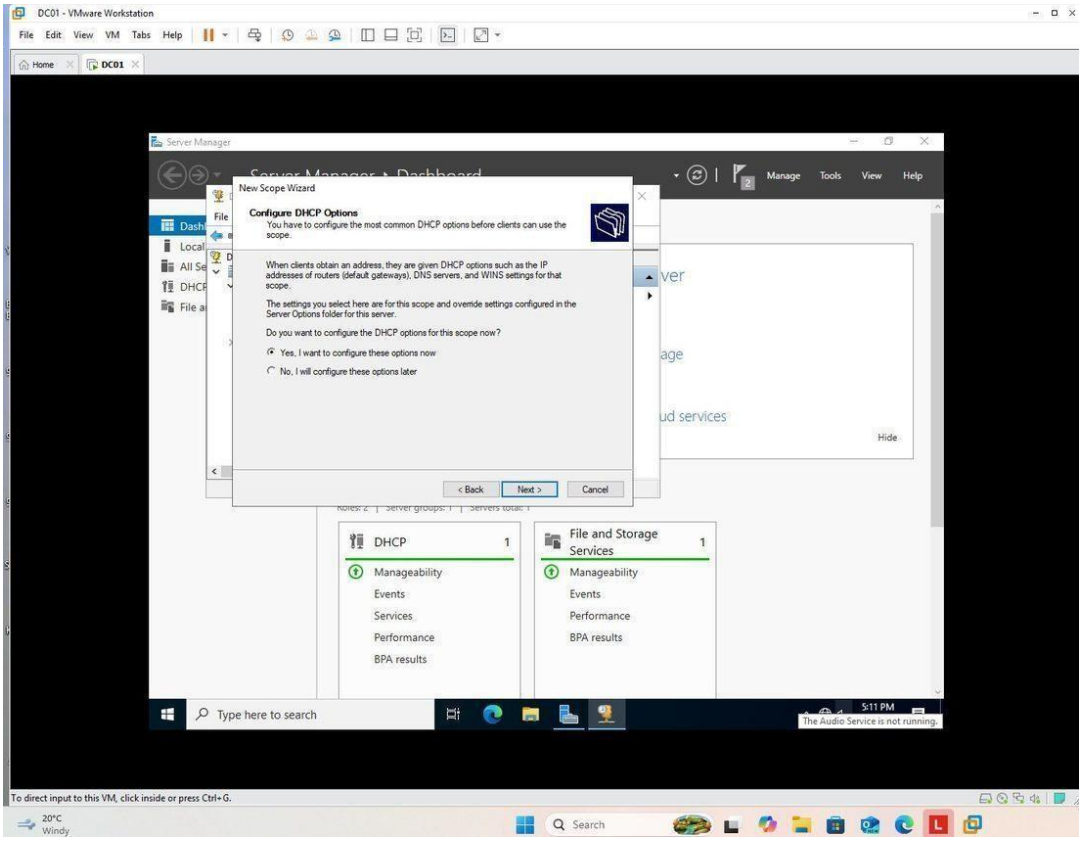


Figure 14: Configure DHCP Options

Step 16: Next was clicked on the router (default gateway) page. Normally the router or gateway would be configured here so that all traffic is managed, but in this lab both machines are on the same LAN segment, so it was not needed.

Step 17: On the Domain Name and DNS Servers page, the parent domain Ethiopia.et and the DNS server 192.168.1.200 were listed, and Next was clicked.

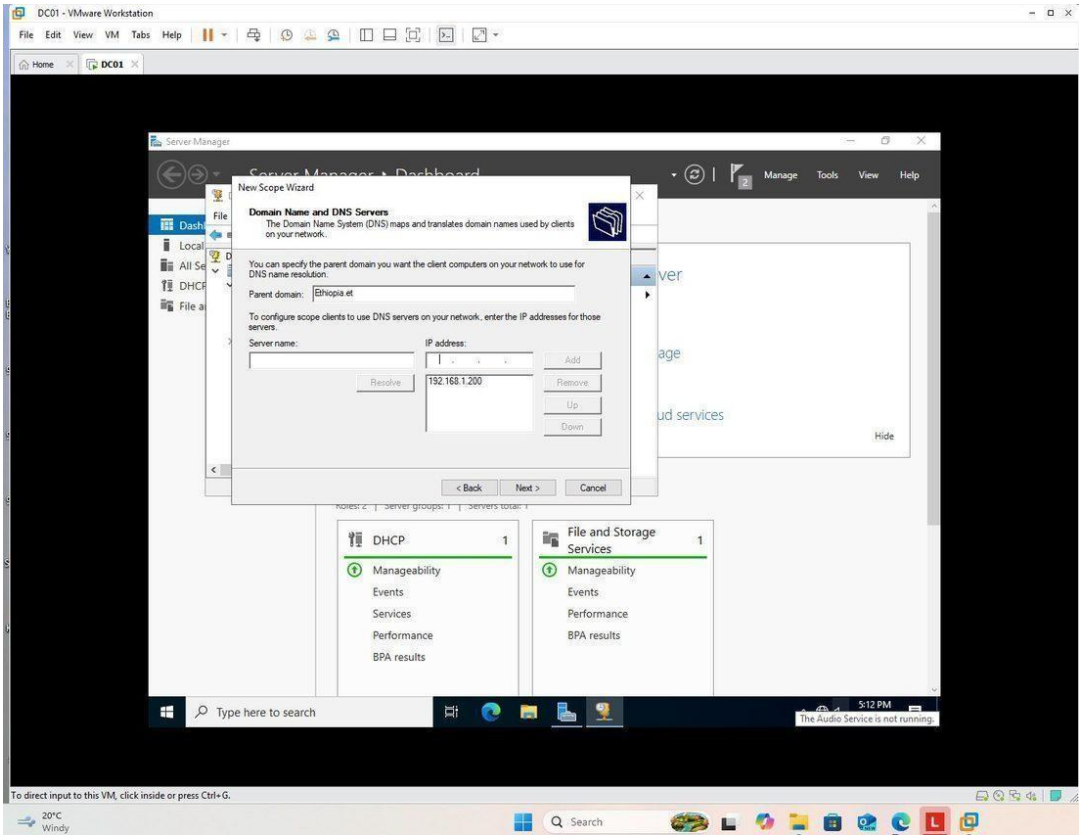


Figure 15: Domain name and DNS server settings

Step 18: Next was clicked on the WINS page, and "Yes, I want to activate this scope now" was selected.

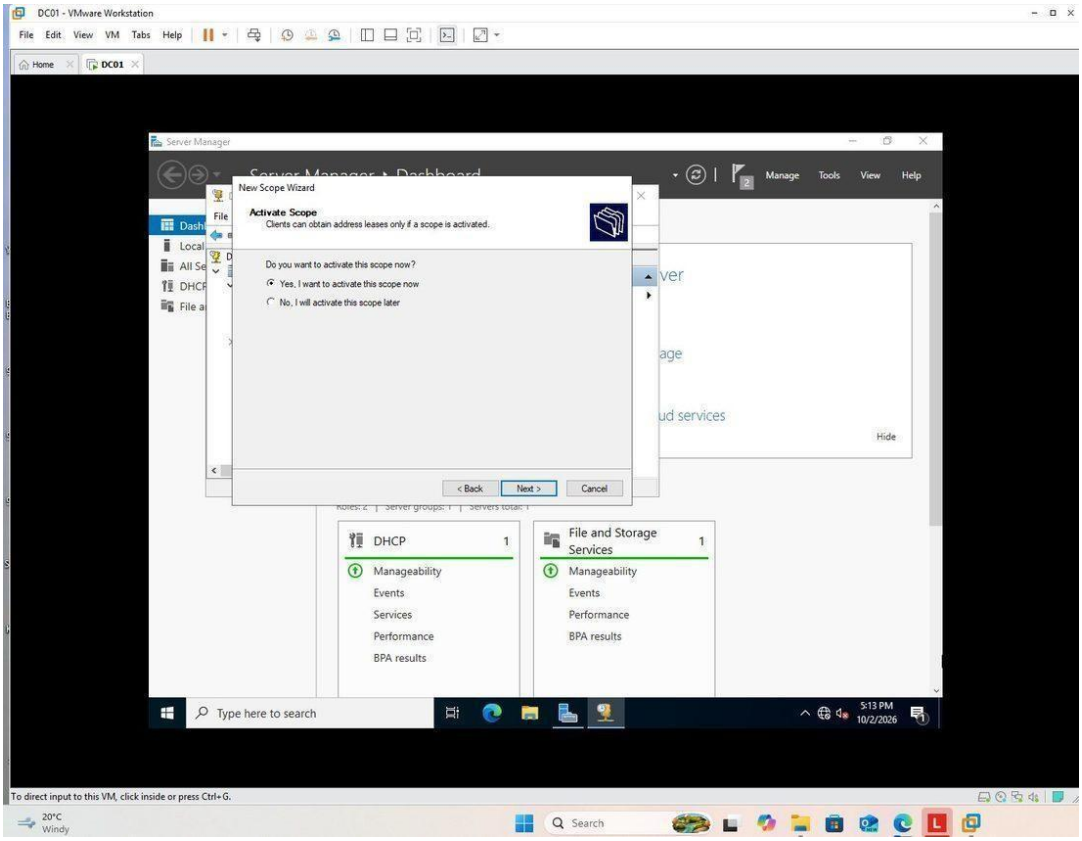


Figure 16: Activate Scope

Step 19: Finish was clicked and the New Scope Wizard was completed.

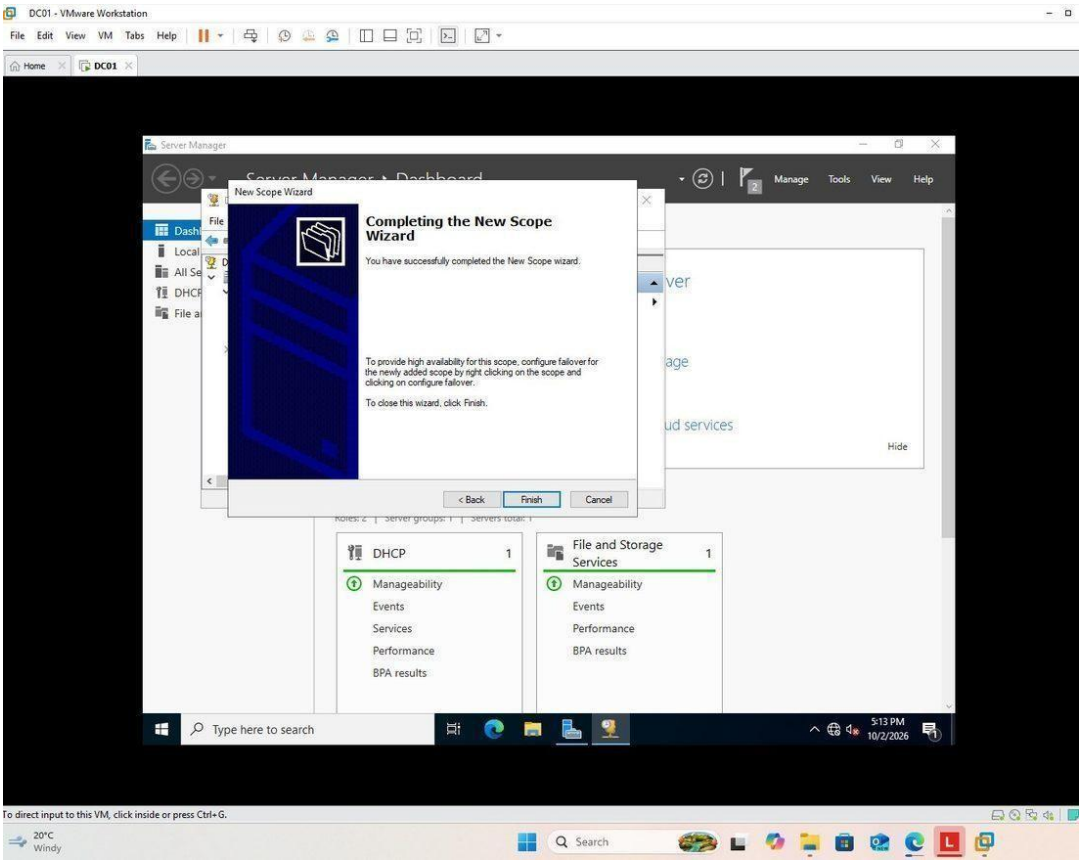


Figure 17: Completing the New Scope Wizard

3.3 Verifying the scope

The new scope now appears in the DHCP console. Under Address Pool, the distributed range 192.168.1.1 to 192.168.1.254 is shown together with the excluded range 192.168.1.1 to 192.168.1.10. The lease time of 1 hour was set in the wizard (Figure 13).

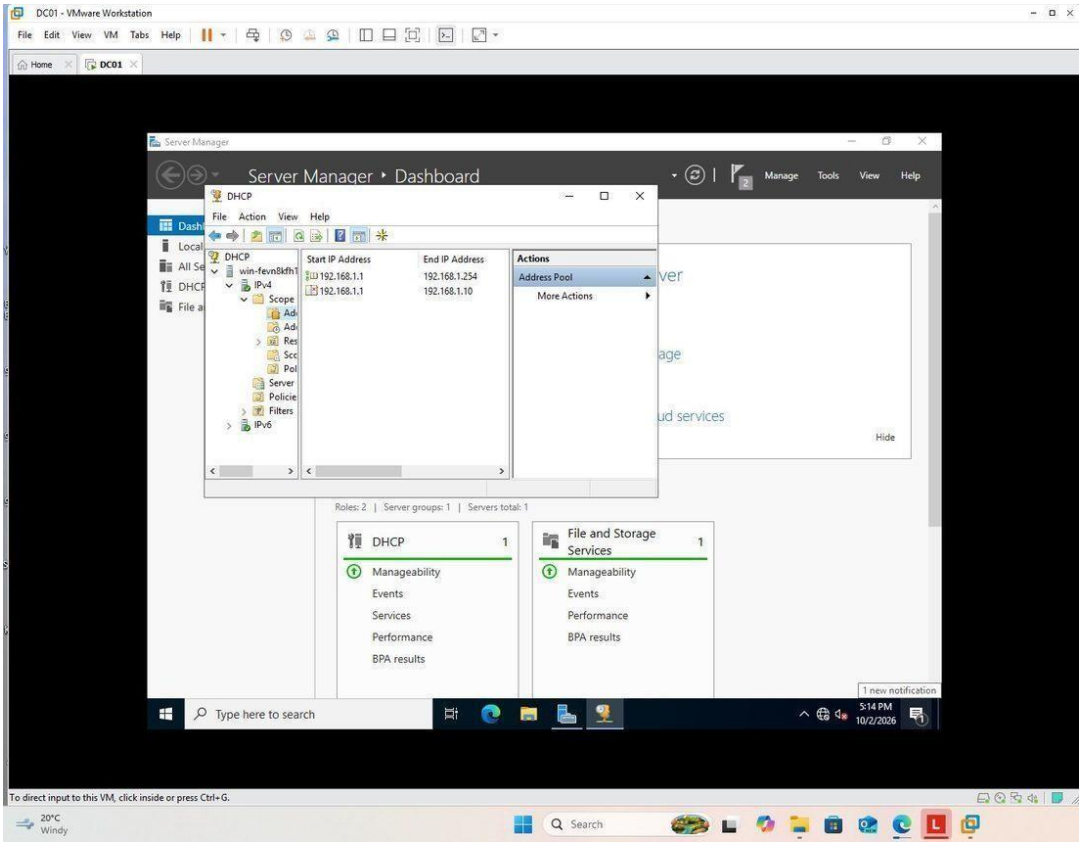


Figure 18: Address Pool showing the scope range and the exclusion range

4. Part 2: Windows Deployment Services (WDS)

For the second part of the lab, WDS is installed to deploy operating systems over the network. DHCP is still required for WDS to function, because WDS uses the Pre-Boot Execution Environment (PXE), which relies on DHCP for IP addressing.

4.1 Installing the WDS role

Step 1: Add Roles and Features was opened again and the Windows Deployment Services role was selected. DHCP Server was shown as already installed.

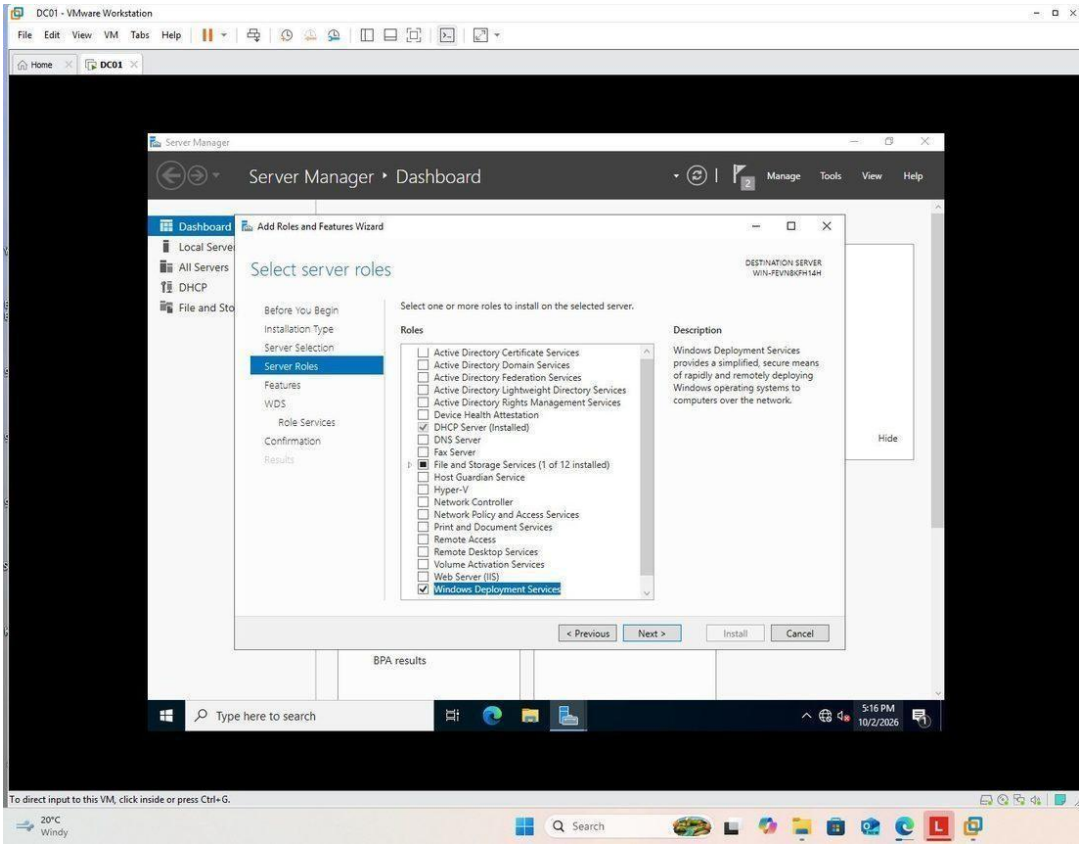


Figure 19: Windows Deployment Services role selected

Step 2: Next was clicked and both role services were kept selected: Deployment Server and Transport Server.

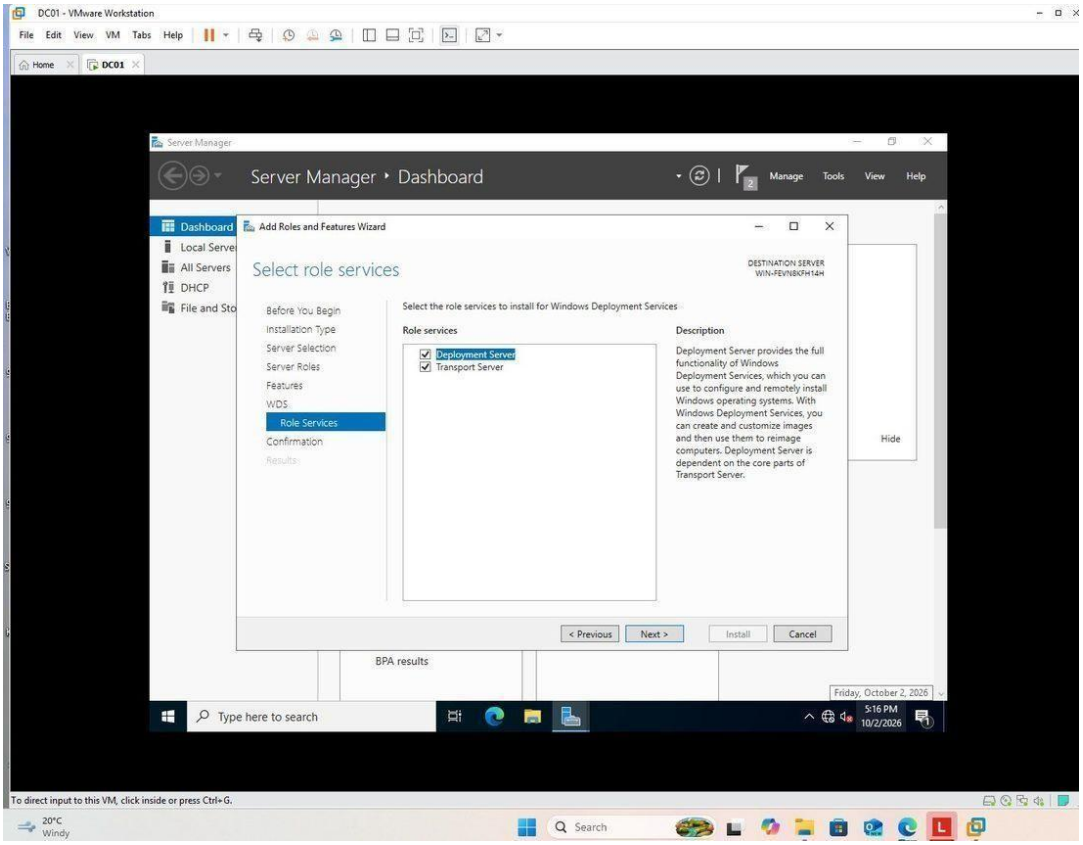


Figure 20: Role services: Deployment Server and Transport Server

Step 3: The confirmation page was reviewed and Install was clicked.

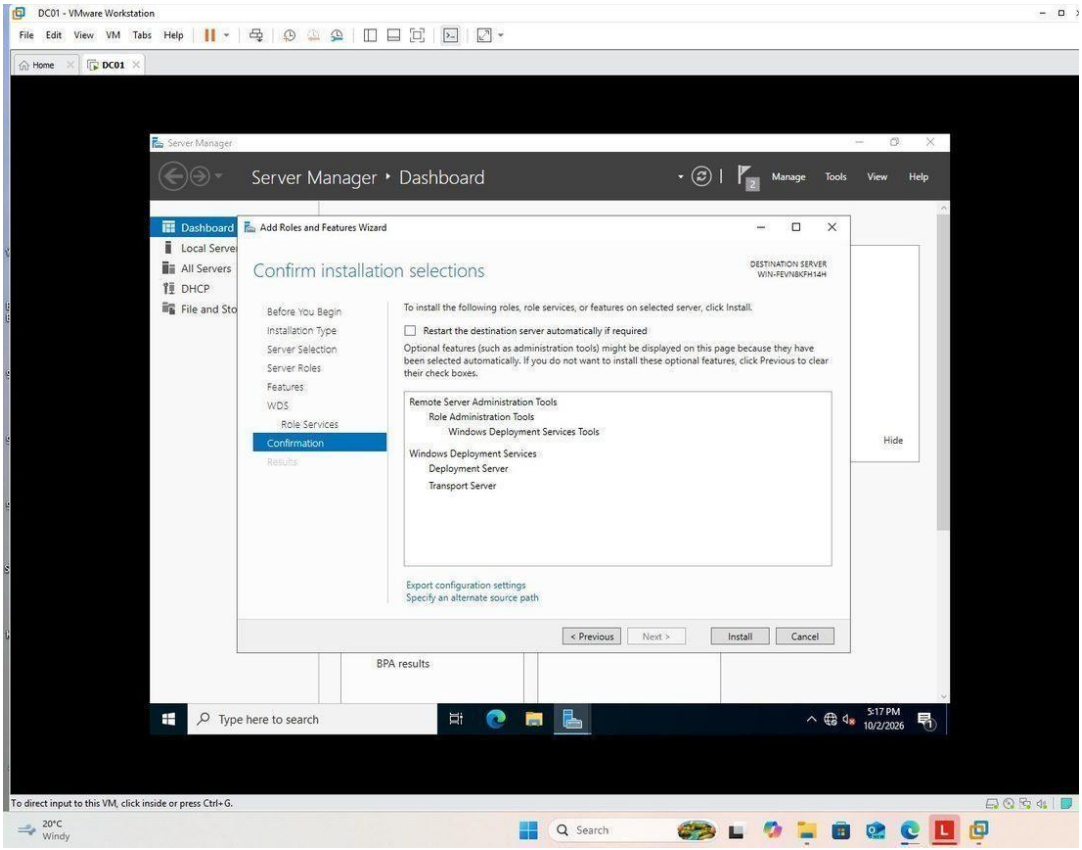


Figure 21: Confirm installation selections for WDS

Step 4: The installation was waited on until it had succeeded. WDS then appeared in the Server Manager menu.

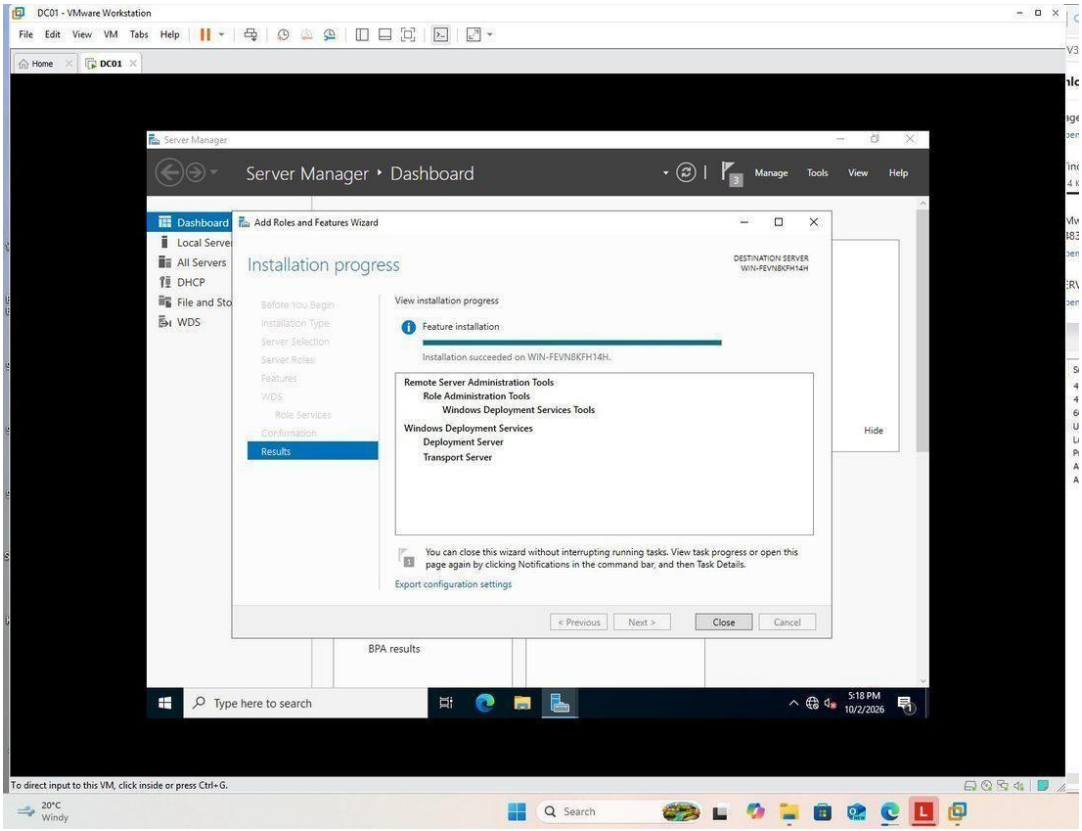


Figure 22: WDS installation succeeded

4.2 Mounting the Windows 10 ISO

Step 5: A Windows 10 ISO image was added to the VM by mounting it in the CD/DVD drive (Virtual Machine Settings). The ISO is needed so that the install.wim and boot.wim image files can be extracted.

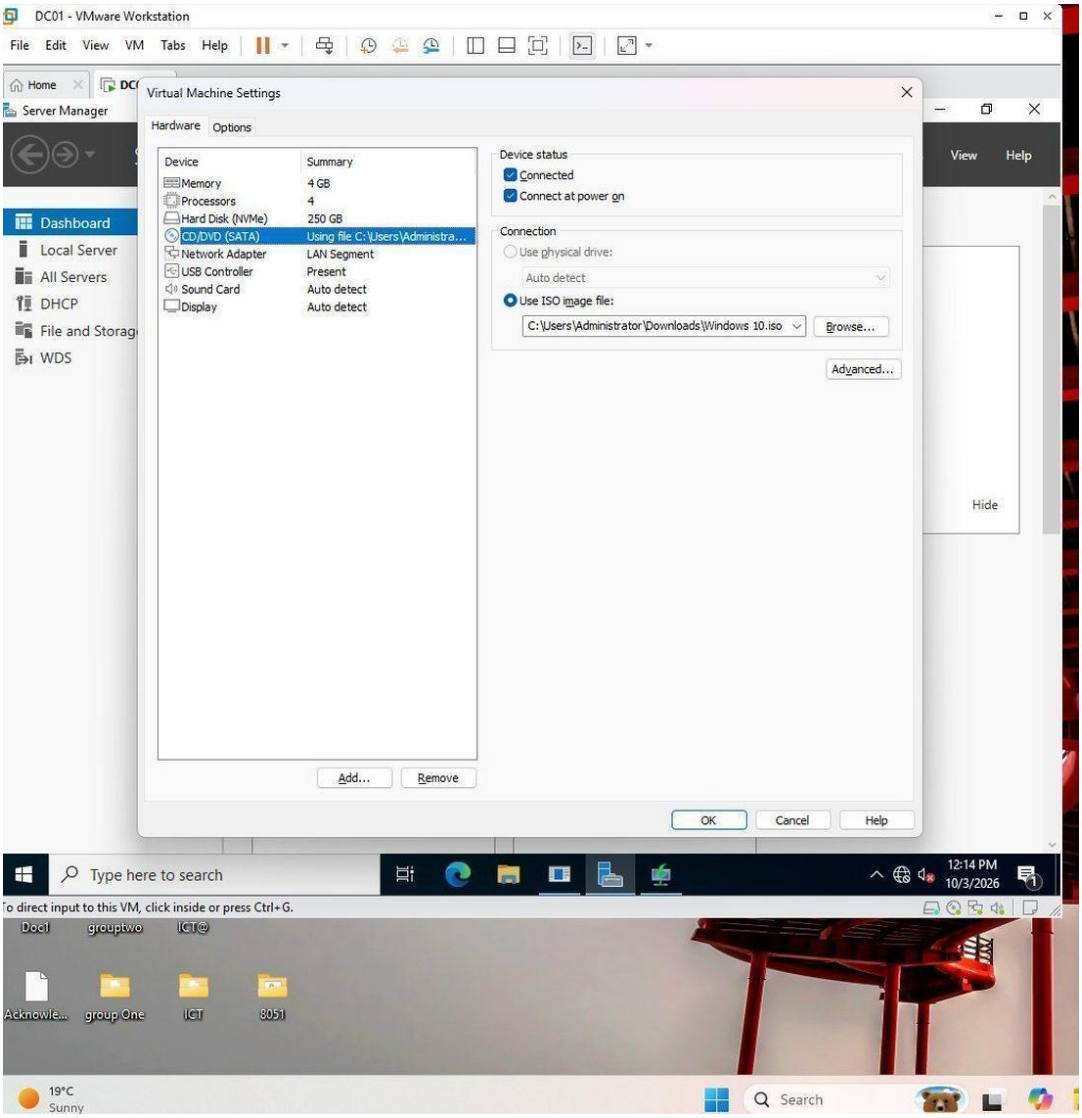


Figure 23: Windows 10 ISO mounted in the CD/DVD drive of DC01

4.3 Configuring the WDS server

Step 6: In Server Manager, Tools was opened and Windows Deployment Services was opened.

Step 7: The server was right-clicked and "Configure Server" was selected.

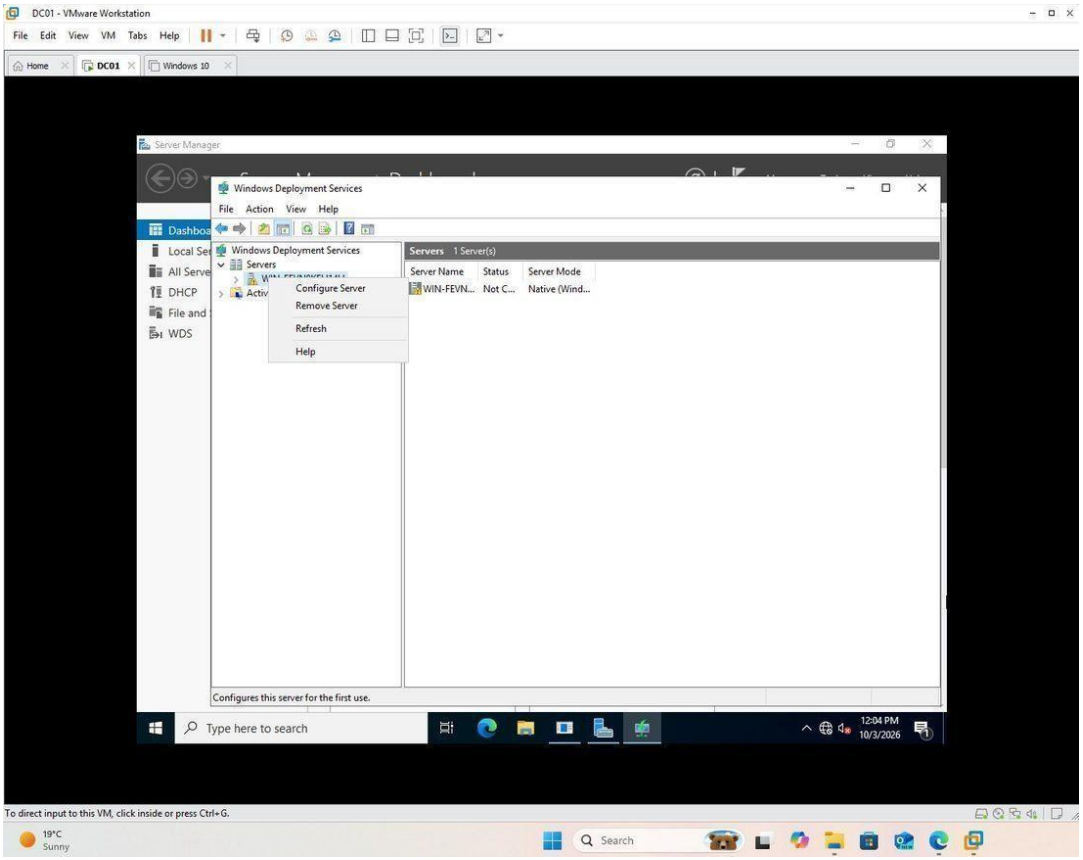


Figure 24: WDS console: Configure Server

Step 8: Next was clicked on the "Before You Begin" page.

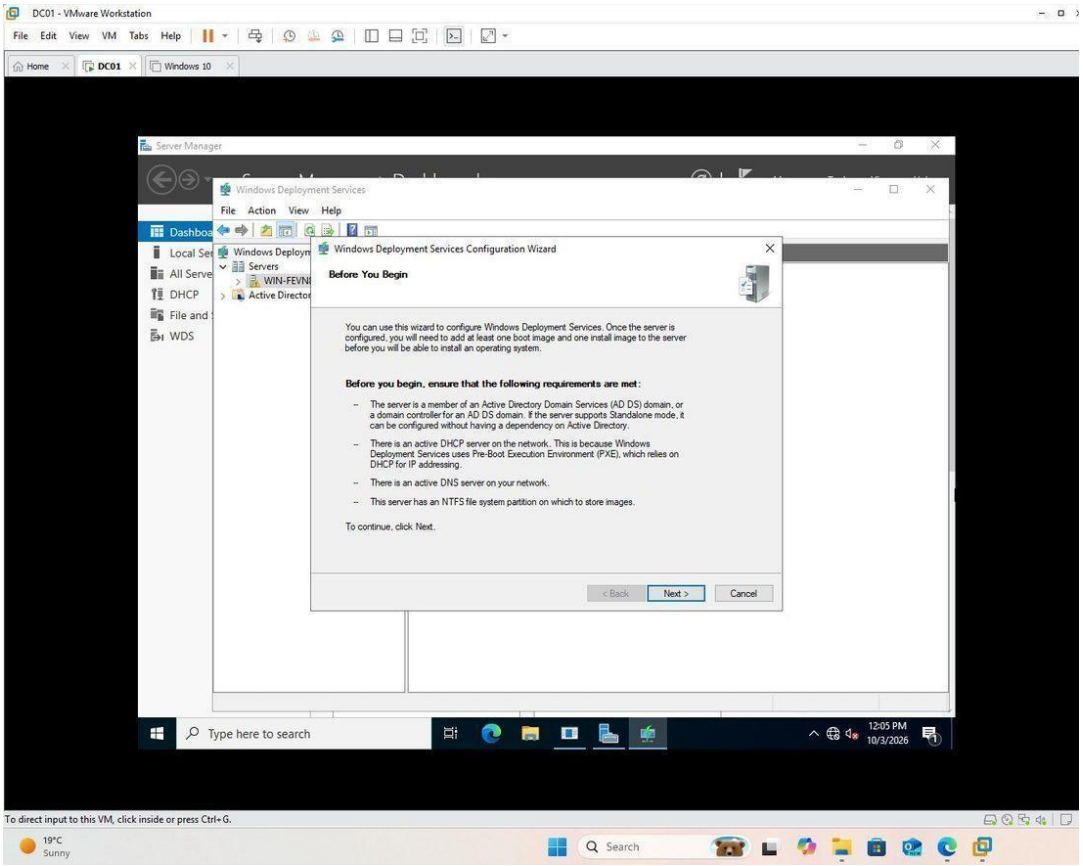


Figure 25: WDS Configuration Wizard: Before You Begin

Step 9: "Integrated with Active Directory" was selected.

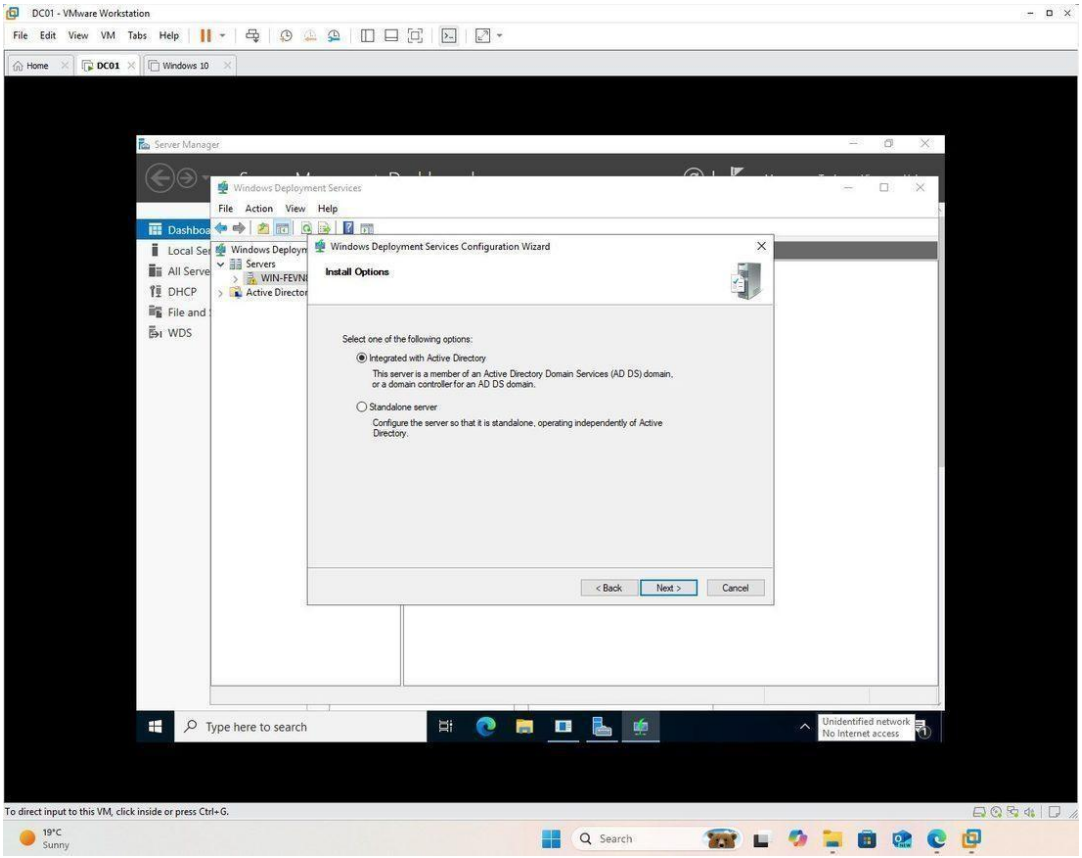


Figure 26: Install Options: Integrated with Active Directory

Step 10: The default Remote Installation Folder (C:\RemoteInstall) was kept. A System Volume Warning was shown because the folder is on the system volume, and Yes was clicked to continue.

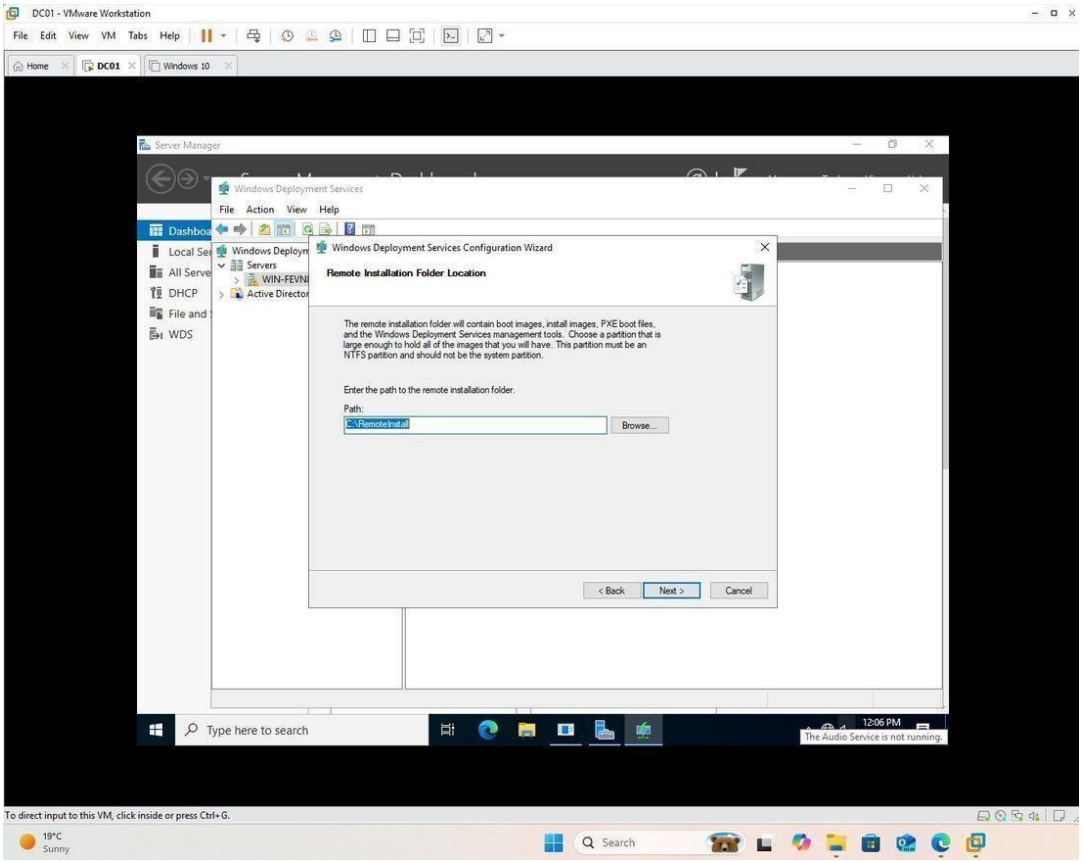


Figure 27: Remote installation folder location

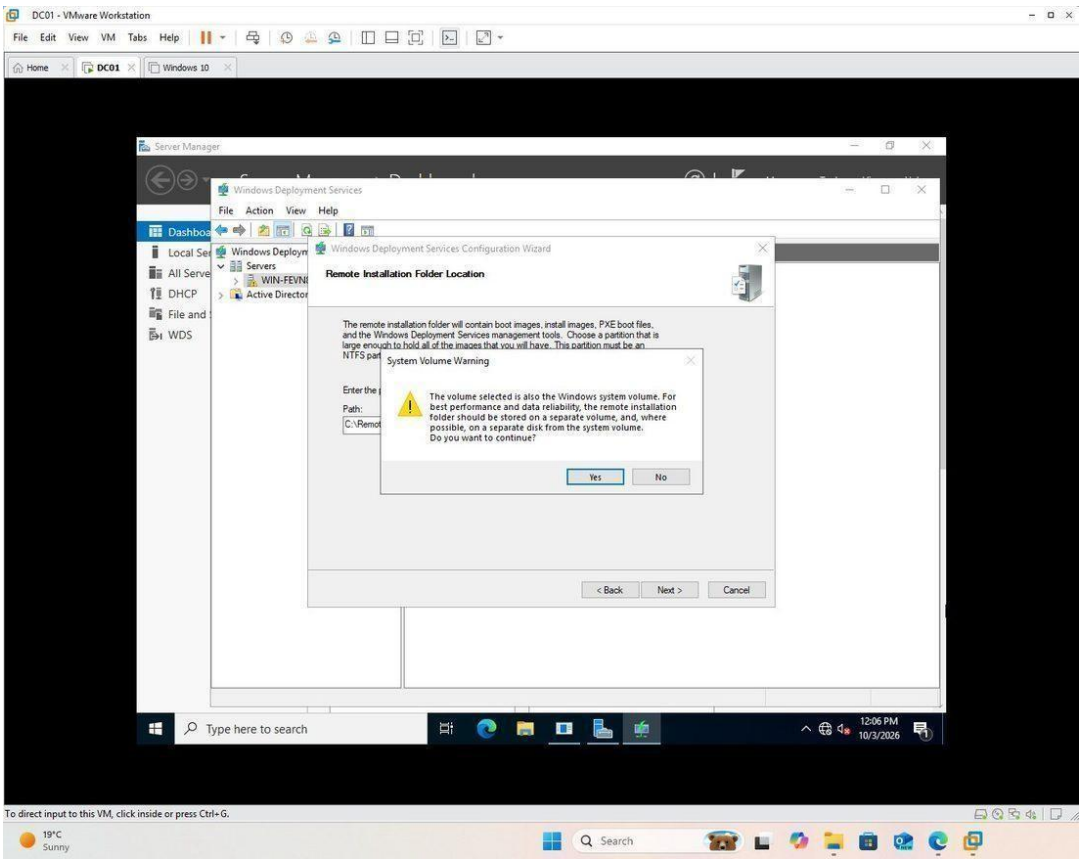


Figure 28: System Volume Warning: click Yes

Step 11: On the PXE Server Initial Settings page, "Respond to all client computers (known and unknown)" was selected.

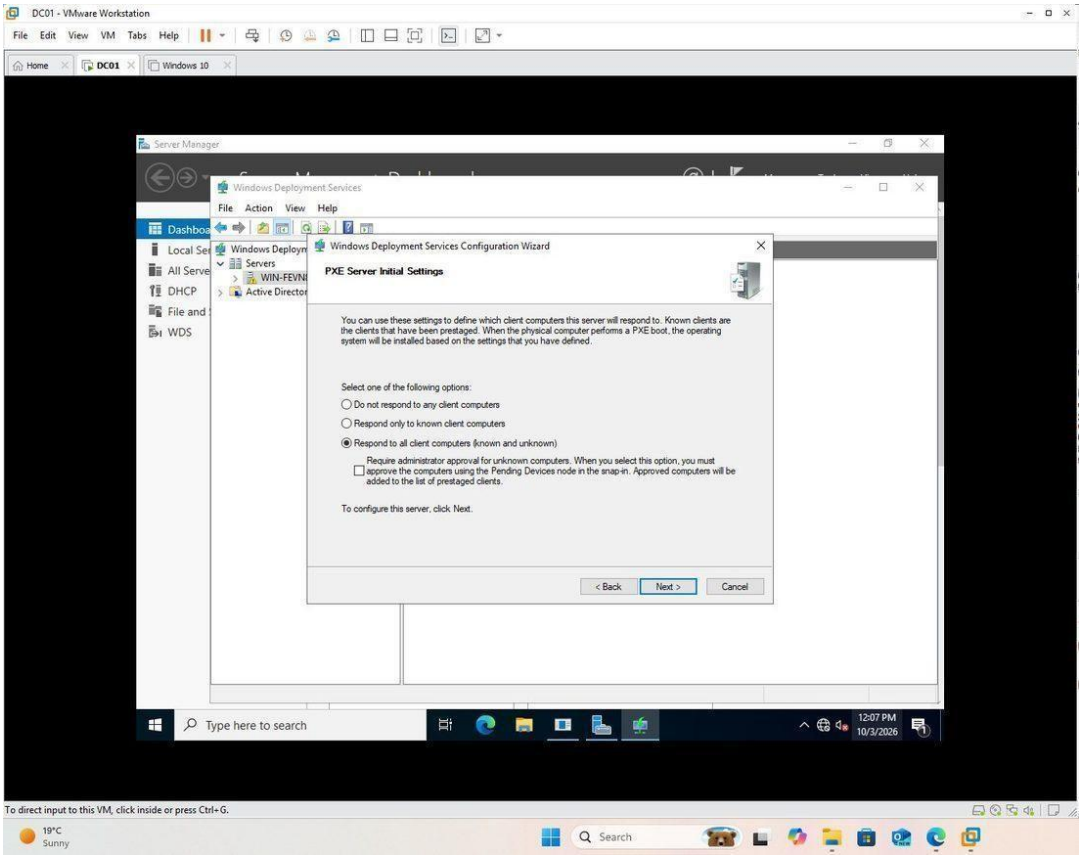


Figure 29: PXE Server Initial Settings

4.4 Troubleshooting: WDS service did not start

At the end of the configuration wizard, the task stopped with the message "The service did not respond to the start or control request in a timely fashion". The WDS server still showed a warning icon in the console.

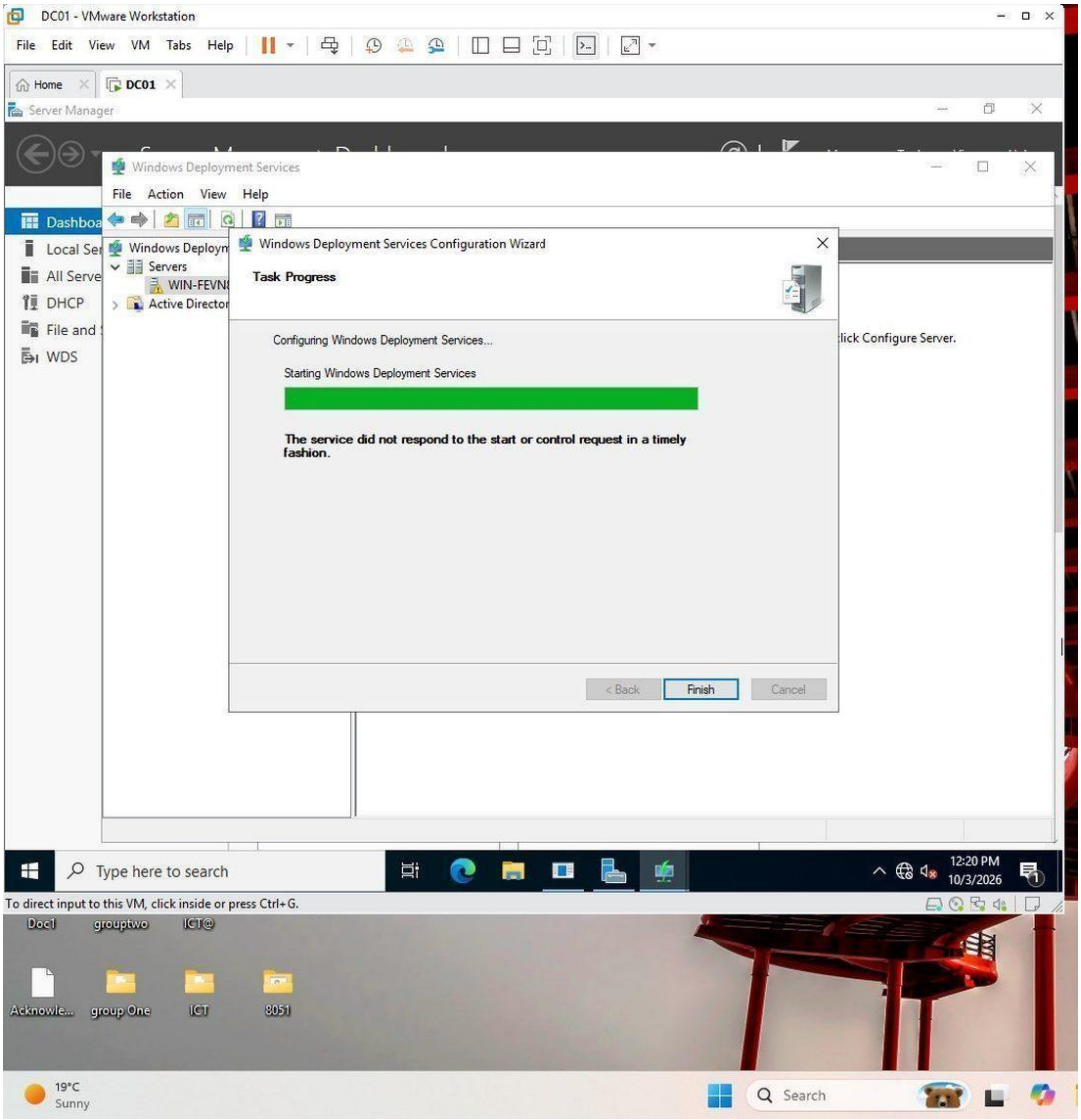


Figure 30: Task progress error during WDS configuration

Resolution: the WDS service was started manually. In the WDS console, the server was right-clicked, then All Tasks and Start were selected. A message confirmed "Successfully started Windows Deployment Services", and the full WDS tree (Install Images, Boot Images, Pending Devices, Prestaged Devices, Multicast Transmissions and Drivers) became available.

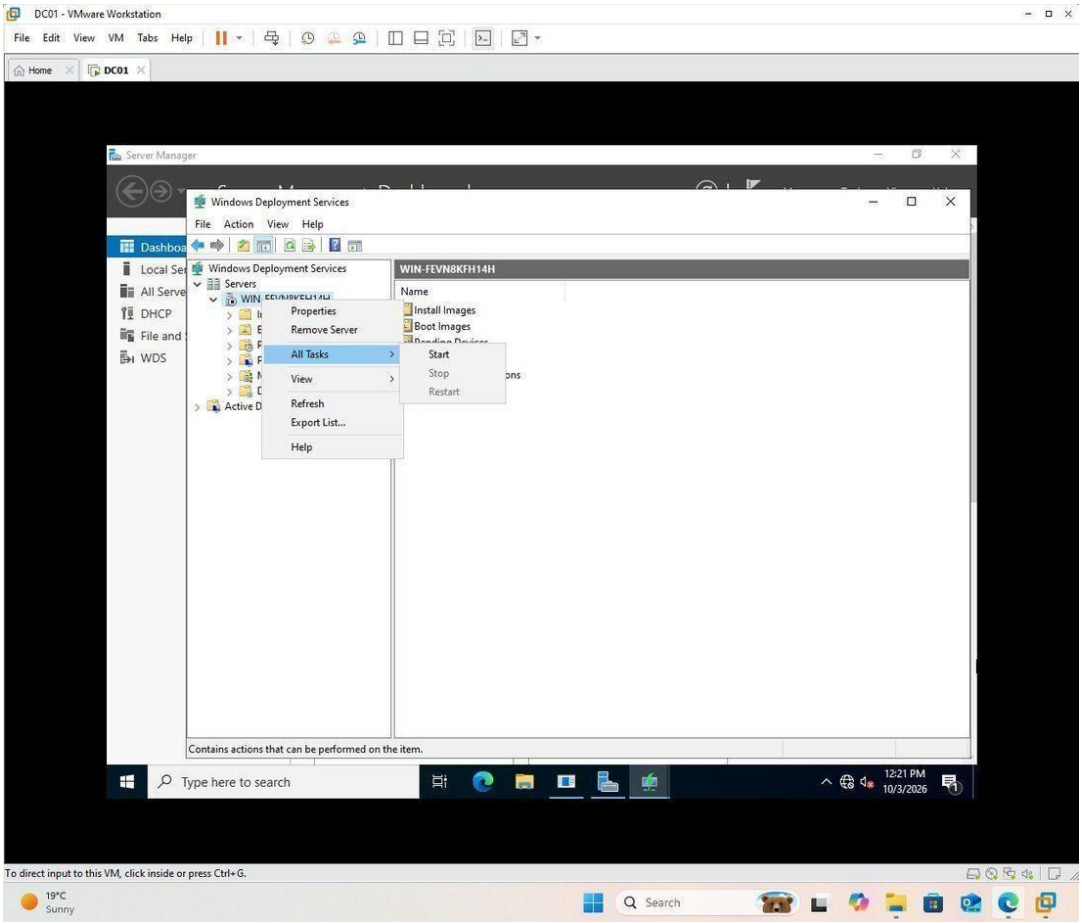


Figure 31: Starting the WDS service: All Tasks, Start

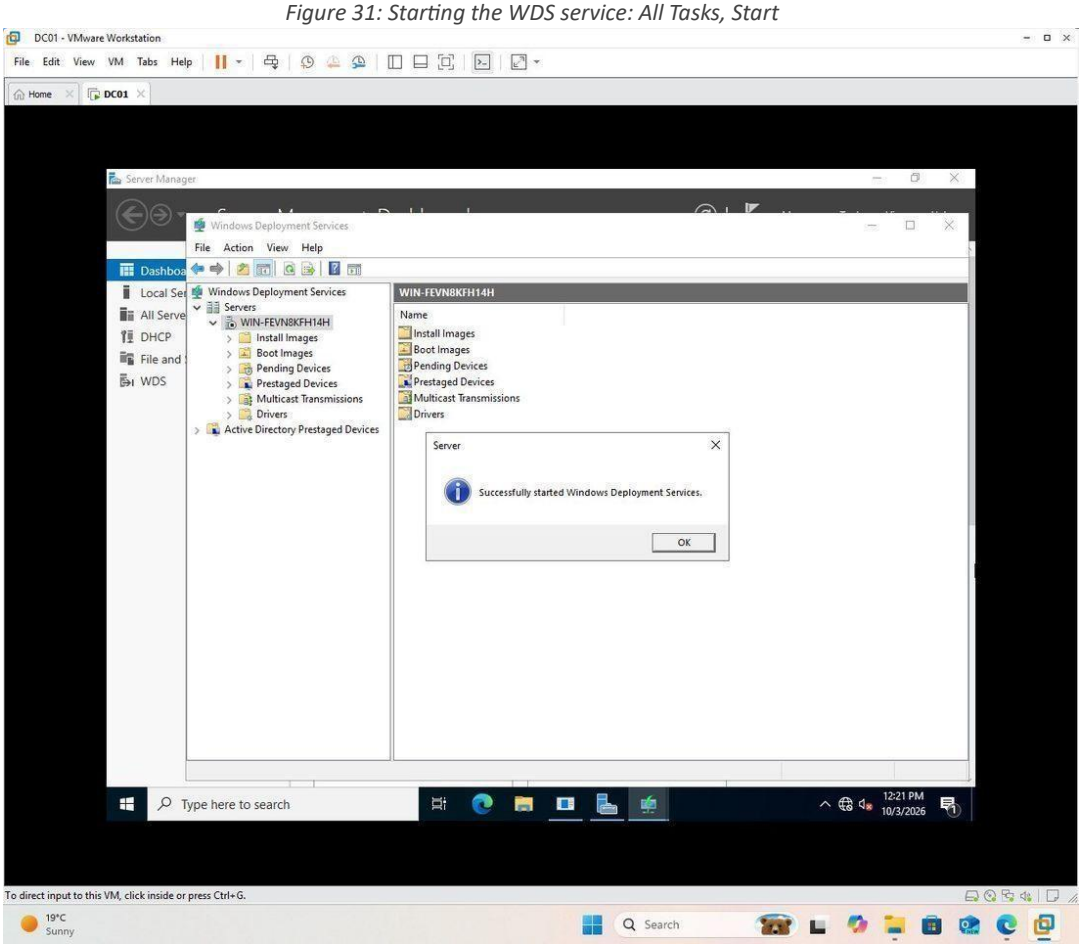


Figure 32: WDS started successfully

4.5 Adding the install image

Step 1: Install Images was right-clicked and "Add Install Image..." was selected.

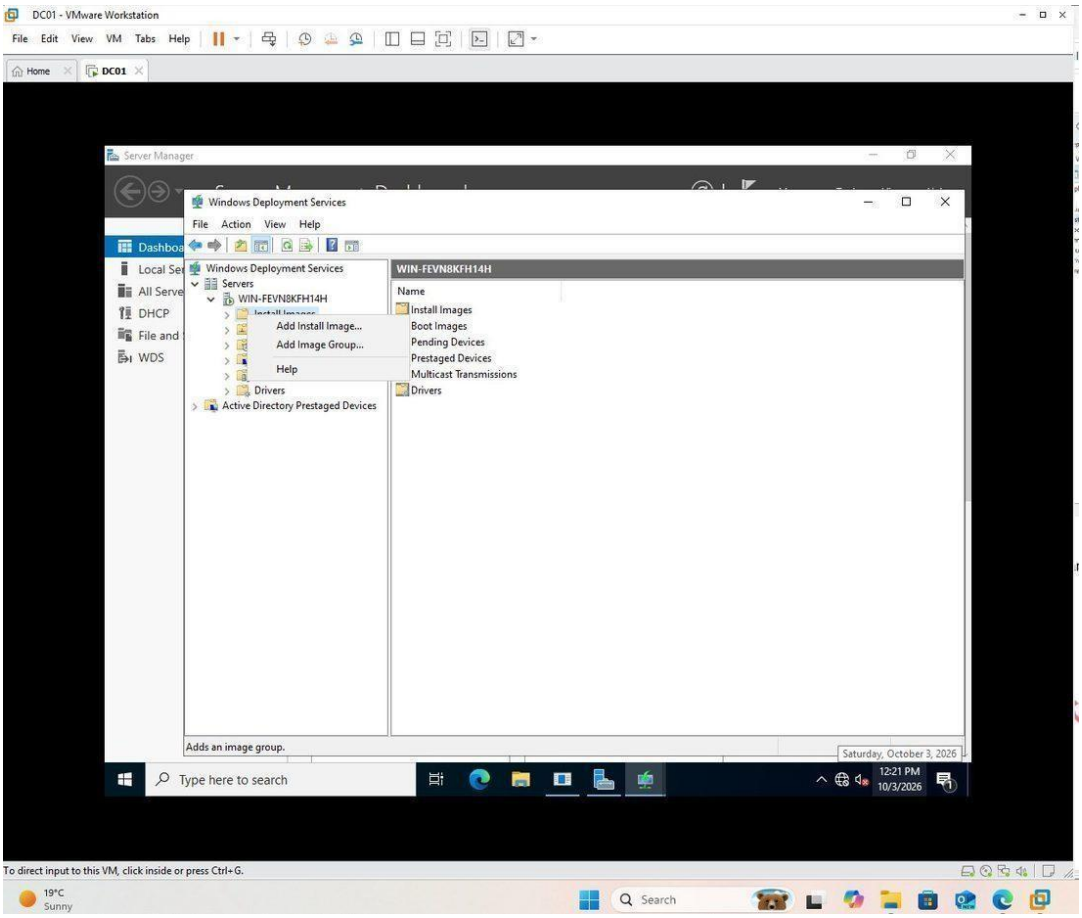


Figure 33: Add Install Image

Step 2: A new image group was created and named "Win 10".

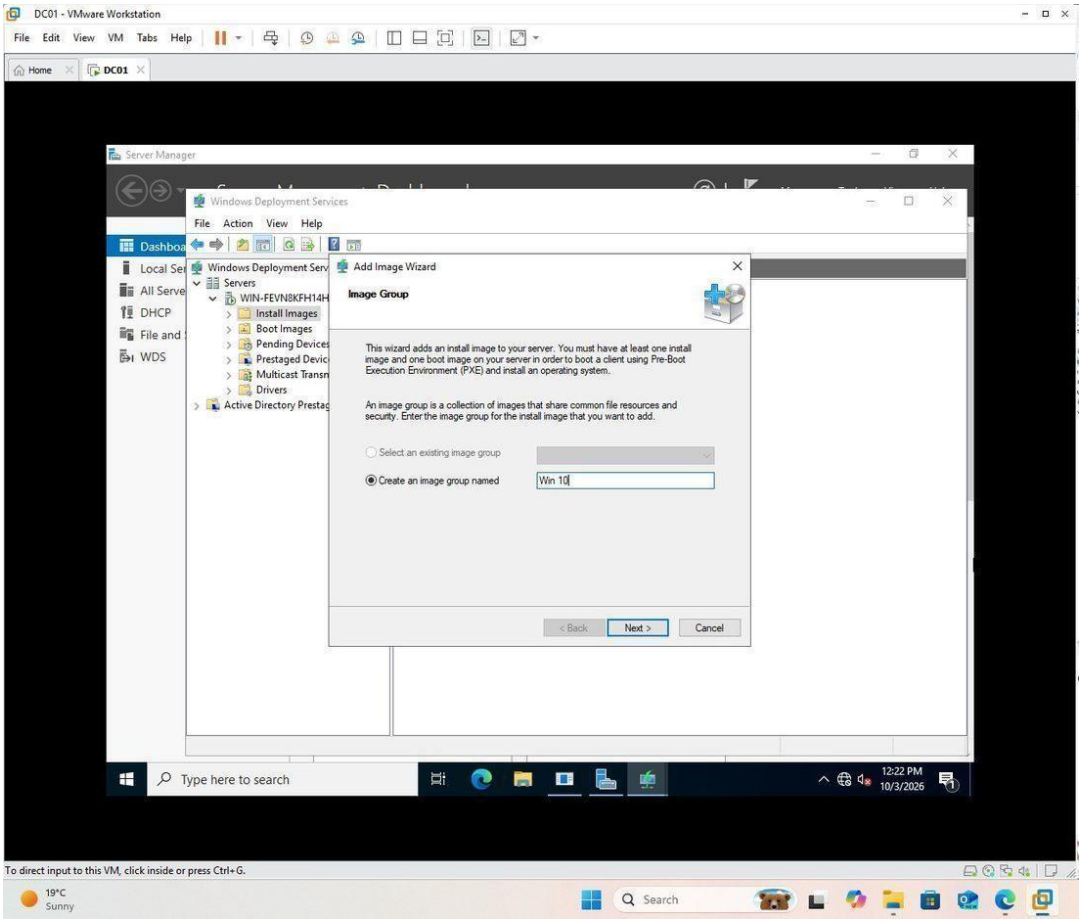


Figure 34: Image group named Win 10

Step 3: The mounted ISO was browsed to and the install image D:\sources\install.wim was selected.

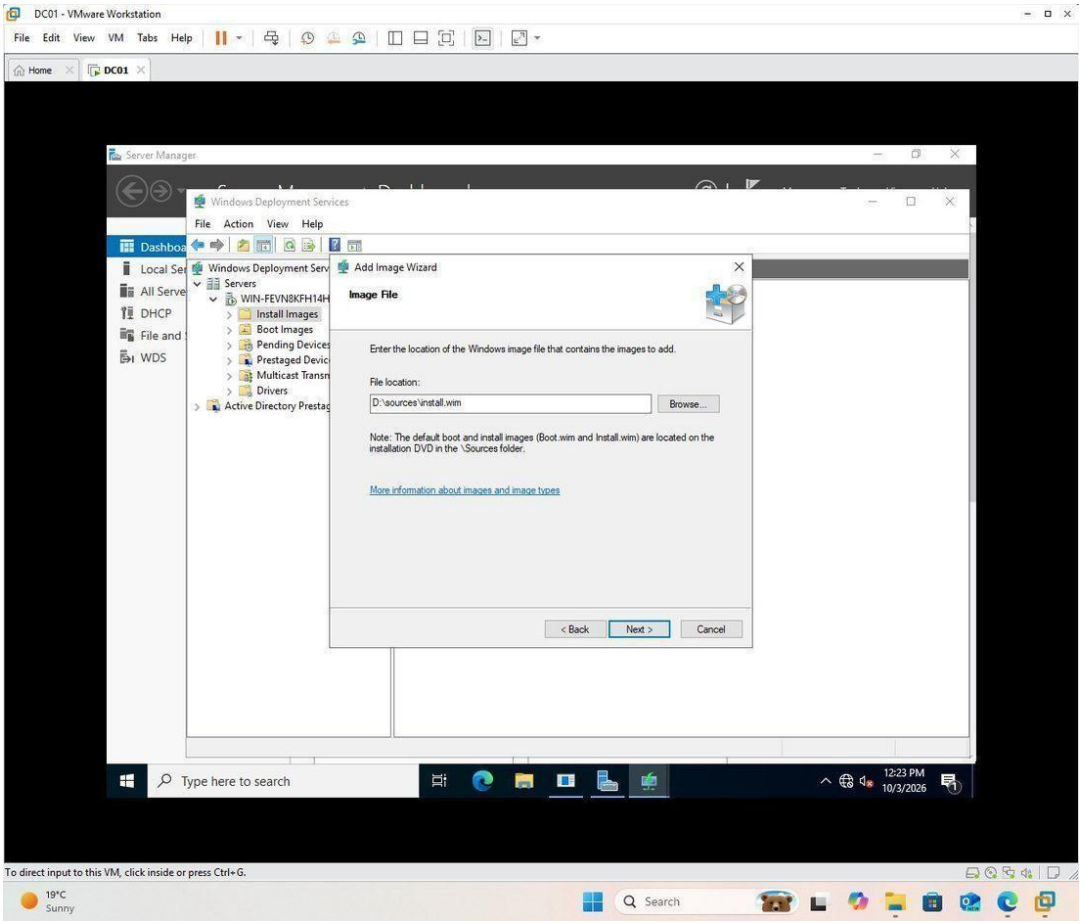


Figure 35: Install image file location

Step 4: The Windows 10 Enterprise Evaluation (x64) image was selected and the default name and description were kept.

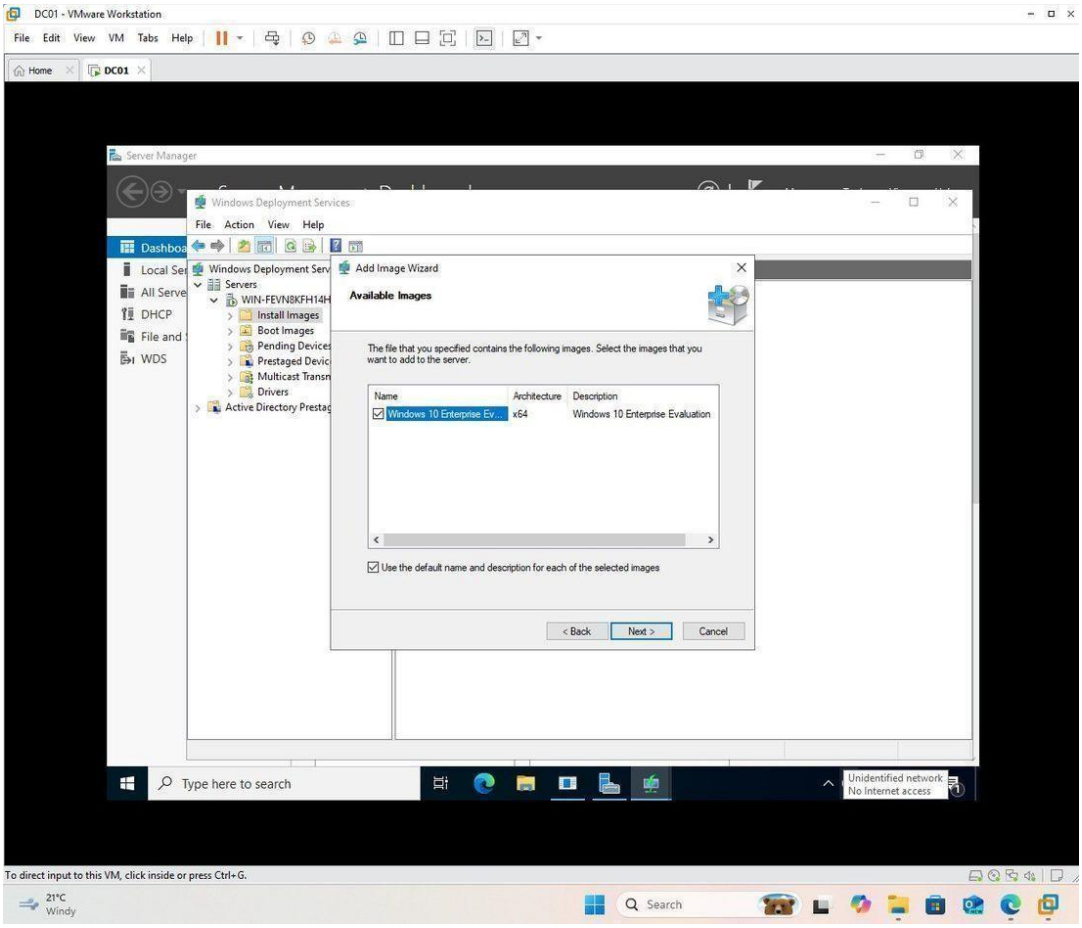


Figure 36: Available images

Step 5: The wizard was completed. The operation was confirmed by the message "The selected images were successfully added to the server".

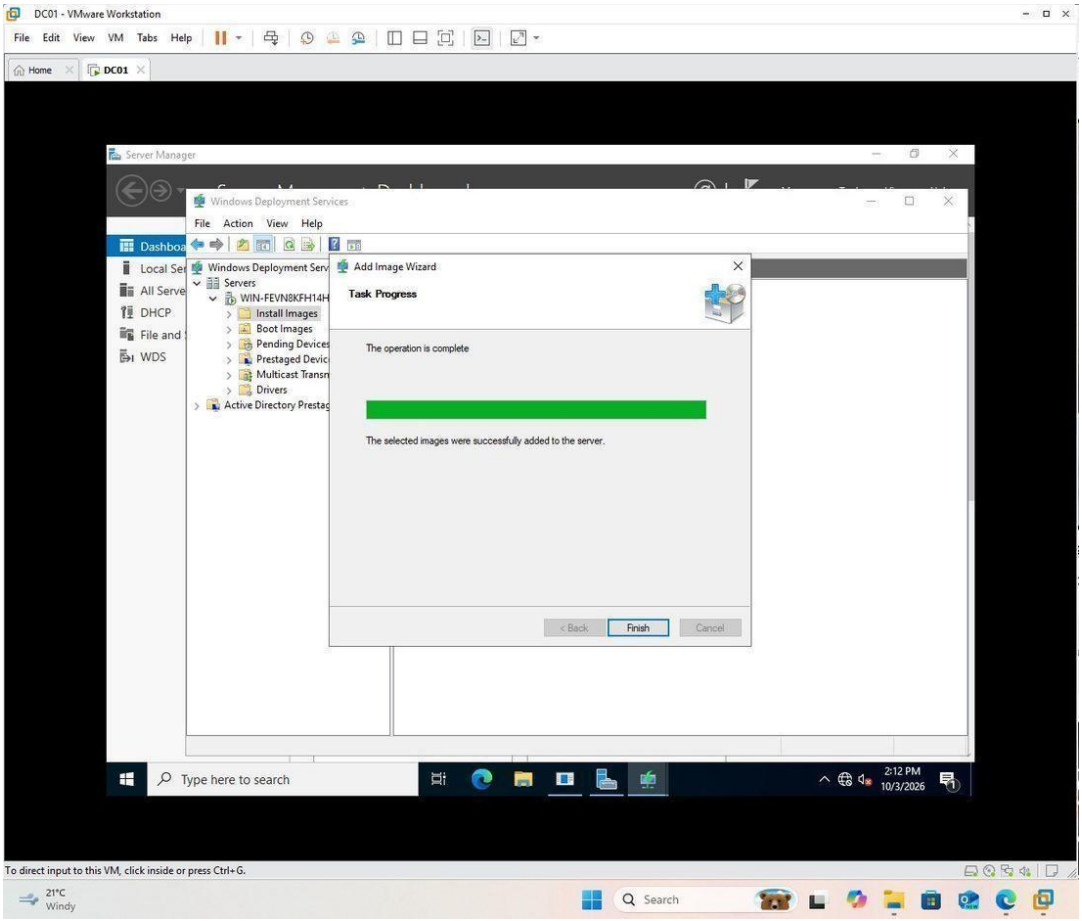


Figure 37: Install image added successfully

4.6 Adding the boot image

A WDS deployment needs two components: an install image for the operating system itself, and a boot image for the booting process. The boot image is found in the same folder of the installation media.

Step 1: Boot Images was right-clicked, "Add Boot Image..." was selected and D:\sources\boot.wim was browsed to.

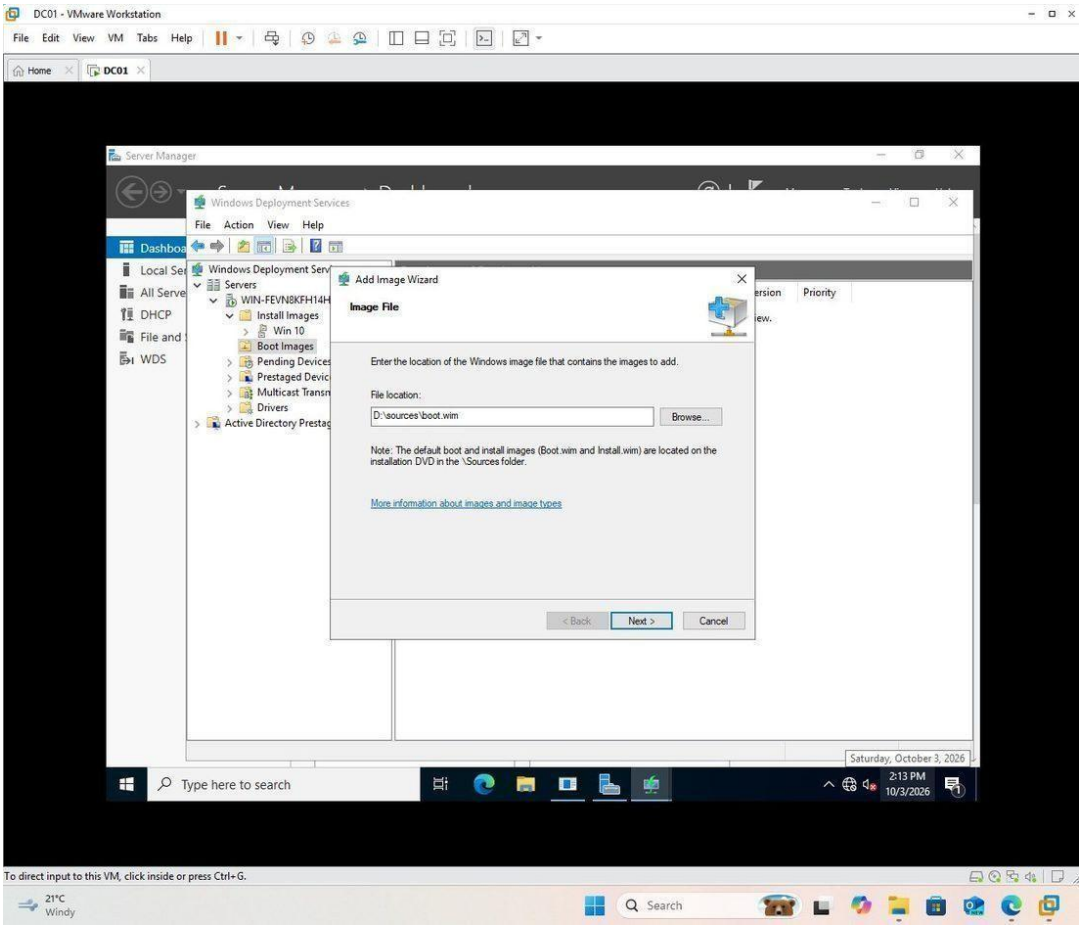


Figure 38: Boot image file location

Step 2: The image name and description "Win 10" were entered.

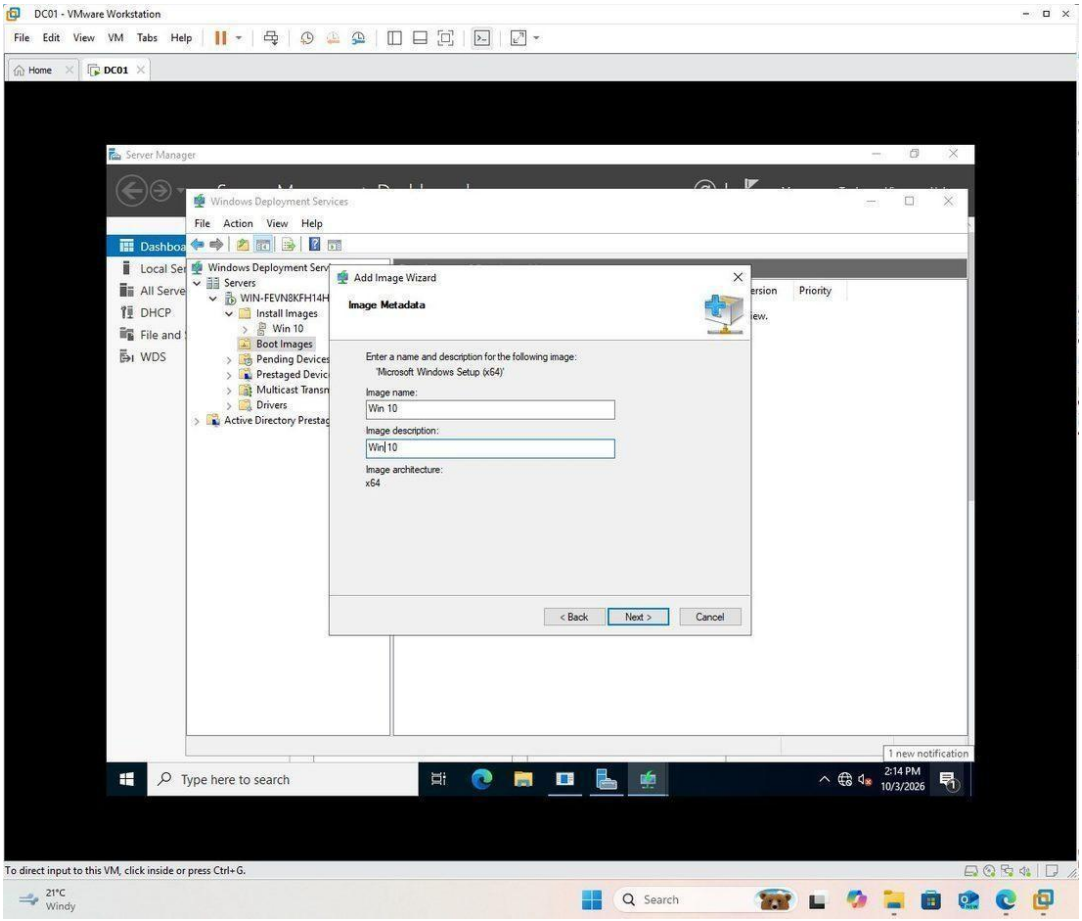


Figure 39: Image metadata

Step 3: The summary was reviewed and Next was clicked.

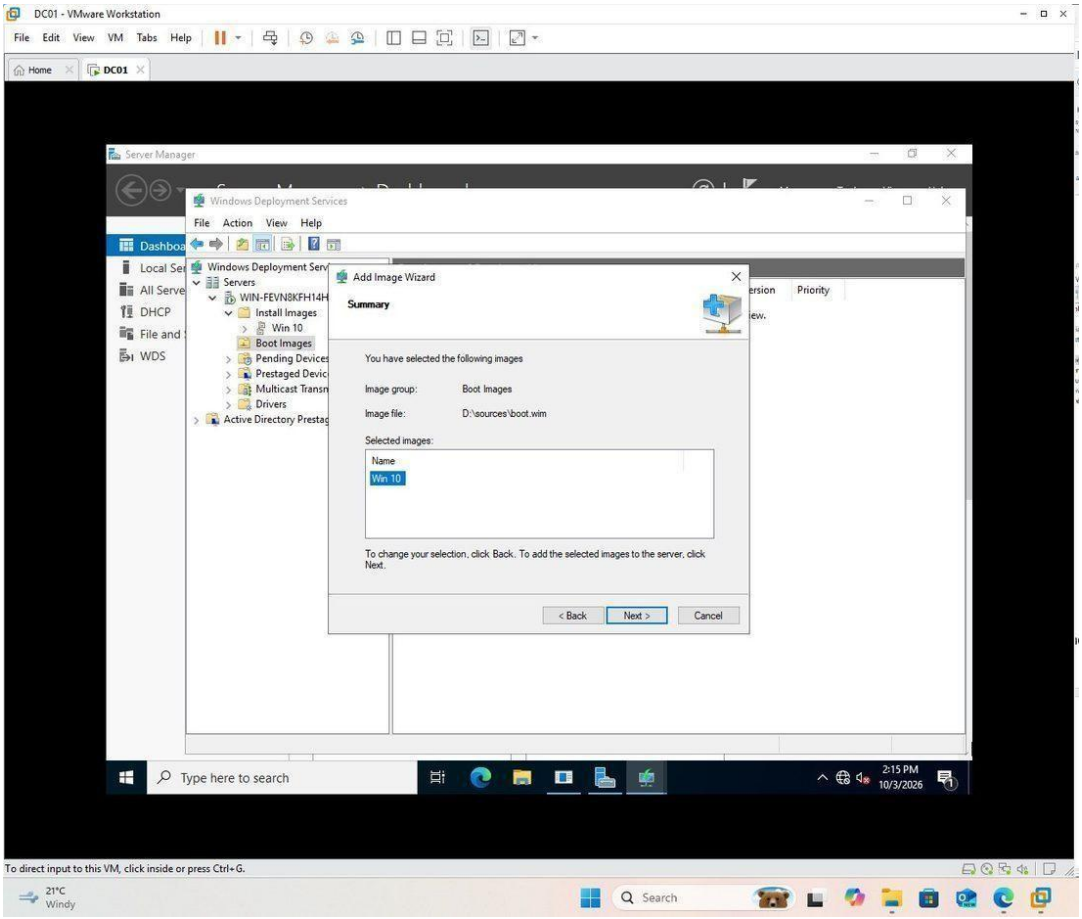


Figure 40: Summary of the boot image to be added

Step 4: The operation was waited on until it had completed and Finish was clicked.

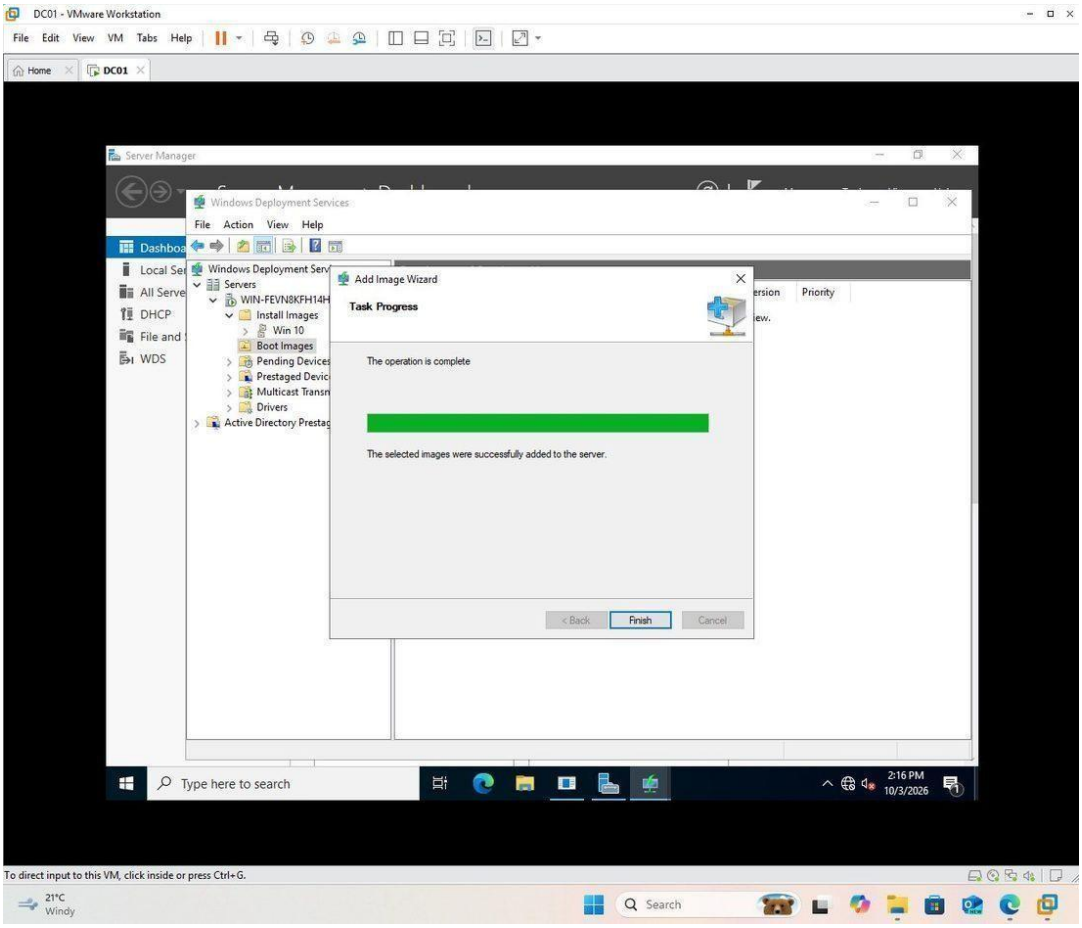


Figure 41: Boot image added successfully

5. Part 3: Deploying Windows 10 over the Network

5.1 Creating the client virtual machine

Step 1: Once the WDS setup had been completed, a new virtual machine named "Win 10 WDS" was created in VMware Workstation.

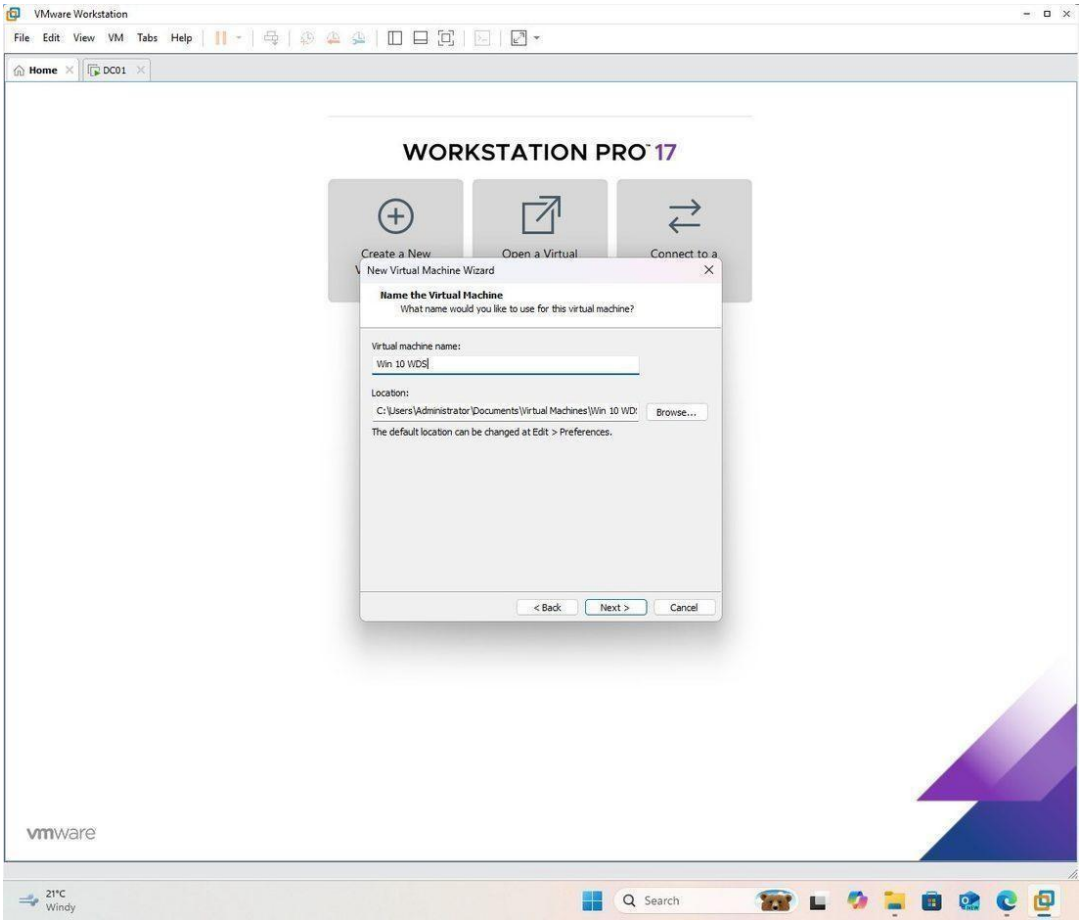


Figure 42: New Virtual Machine Wizard: Win 10 WDS

Step 2: The hardware was configured (4 GB RAM, 4 processors) and the network adapter was set to the same LAN Segment as DC01. No ISO image was added: the CD/DVD was left empty, because the VM must receive the PXE boot signal from WDS.

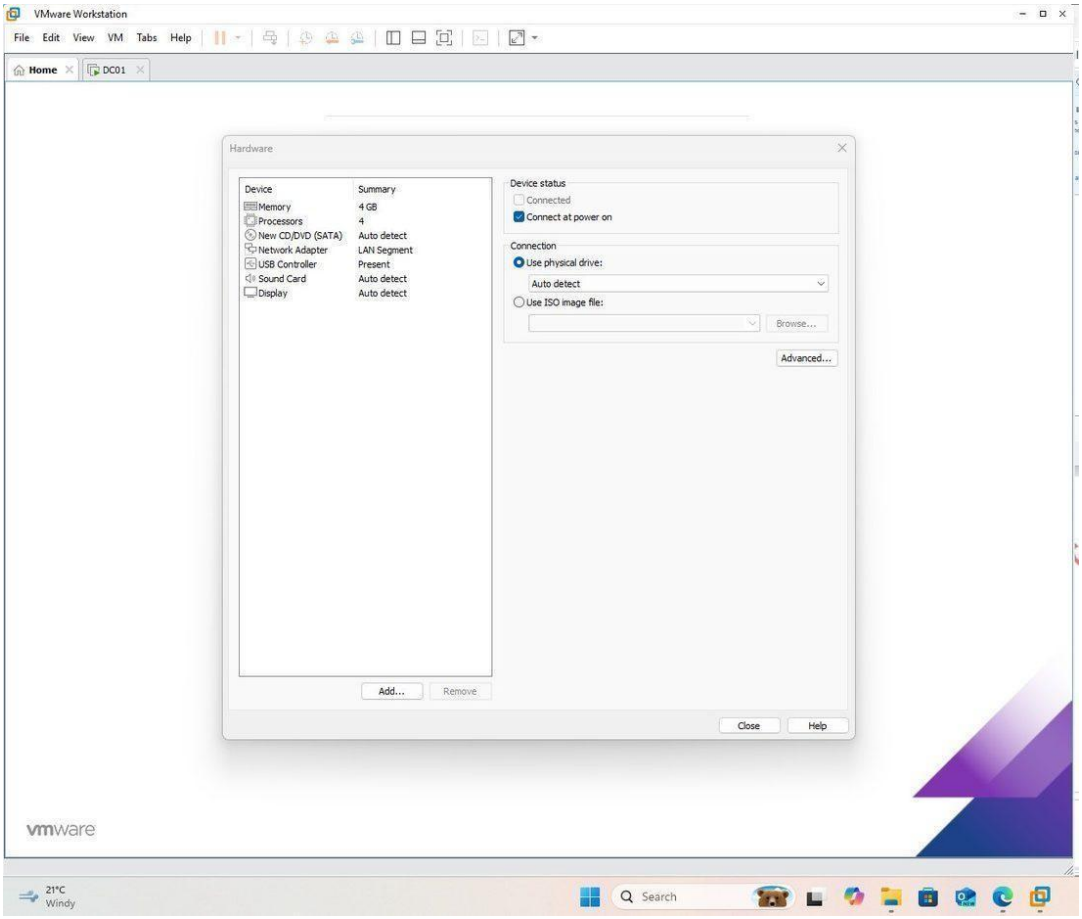


Figure 43: Client VM hardware: network adapter on LAN Segment, CD/DVD empty

5.2 PXE boot and Windows setup

Step 3: The VM was powered on and F12 was pressed so that the PXE boot signal from WDS would continue to be received. If F12 does not work, the Fn key combination (Fn + F12) is used.

Step 4: An IP address was obtained by the client from the DHCP server, WDS was contacted and the boot image was loaded. The Windows Setup screen of Windows Deployment Services appeared, and the locale and keyboard layout were selected and Next was clicked.

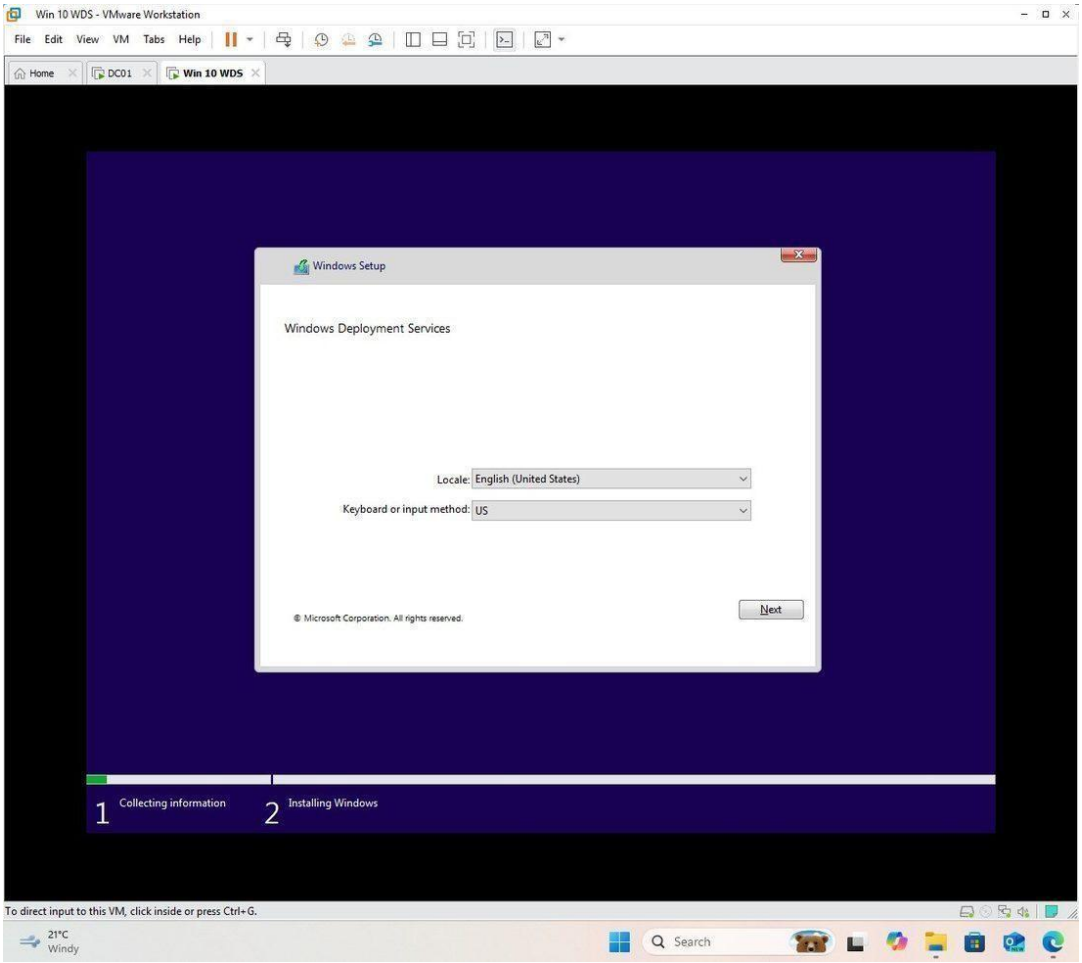


Figure 44: Windows Setup loaded from WDS over the network

Step 5: The credentials for connecting to the WDS server (WIN-FEVN8KFH14H\Administrator) were entered.

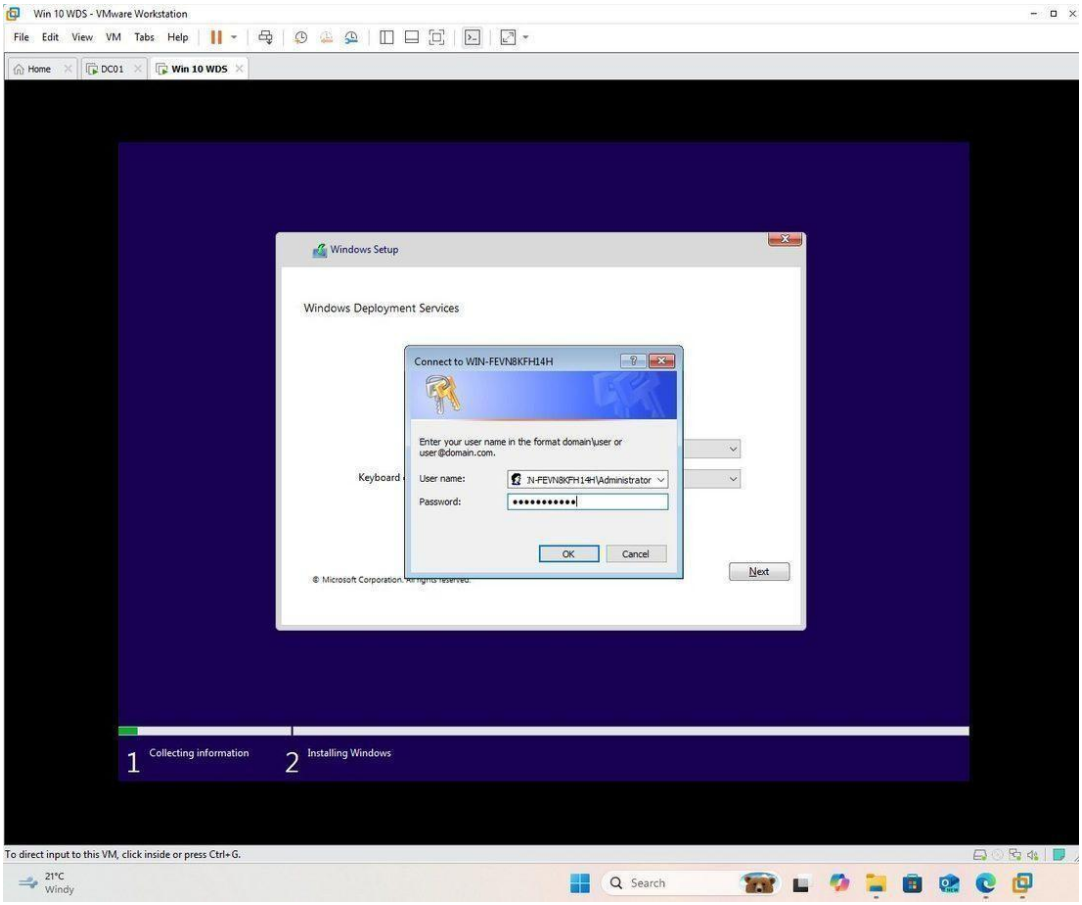


Figure 45: Connecting to the WDS server

Step 6: The operating system to be installed was selected: Windows 10 Enterprise Evaluation (x64), the image that was added earlier.

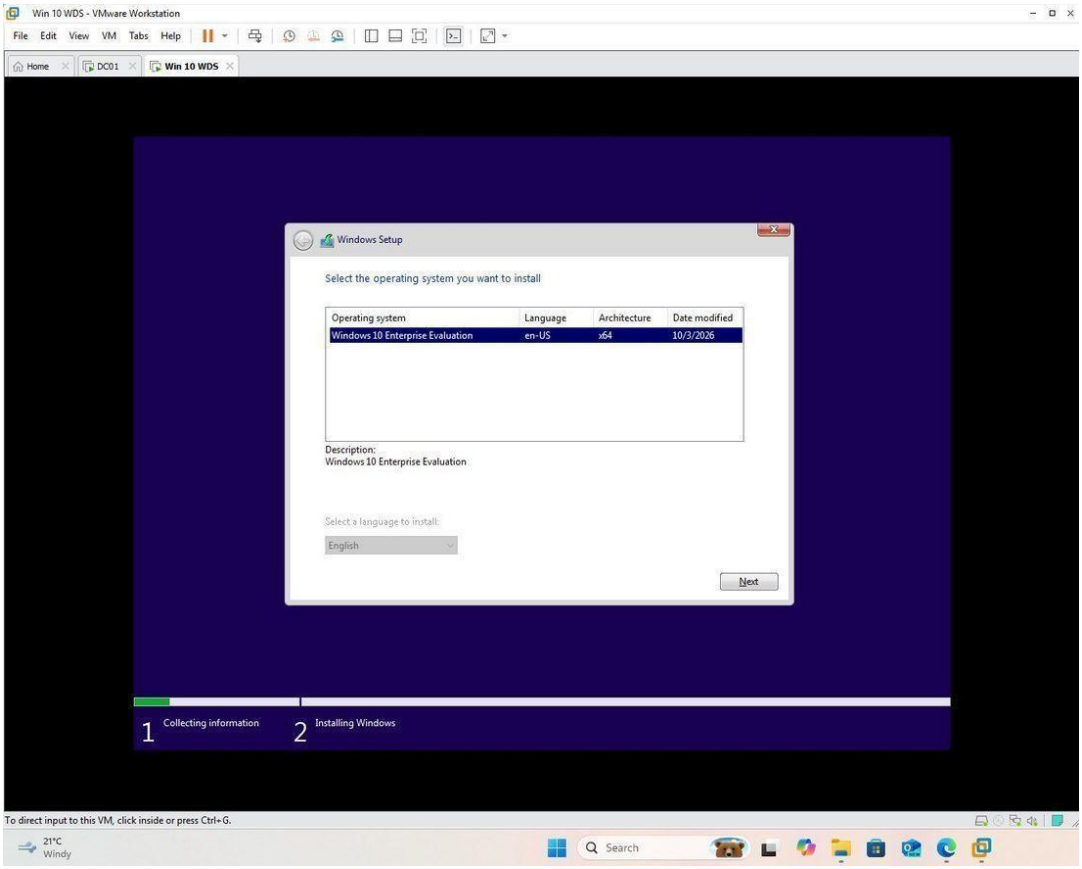


Figure 46: Select the operating system to install

Step 7: The disk (Drive 0, 15.0 GB) was selected and Next was clicked.

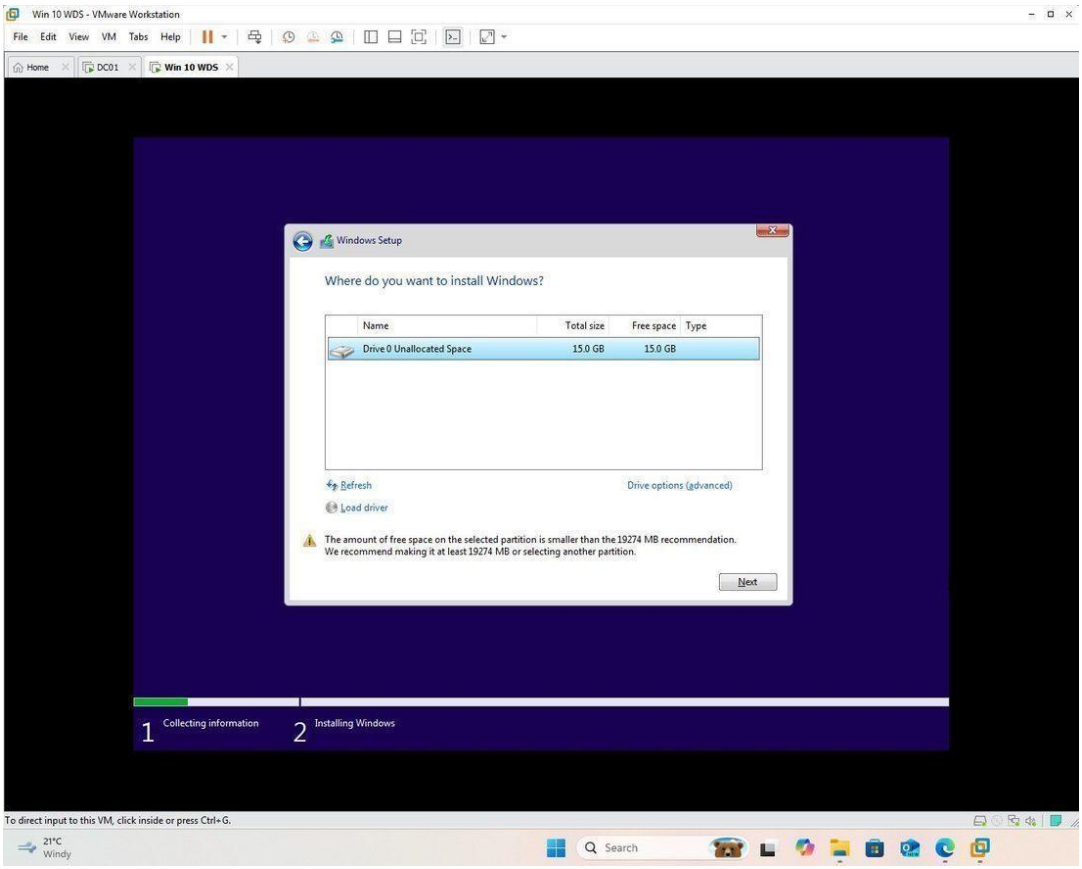


Figure 47: Where do you want to install Windows?

Note: Setup shows a warning that the free space is smaller than the 19,274 MB recommendation. The installation still proceeded. In a production environment, a larger disk should be used.

Step 8: Windows was installed from the server. The computer was restarted several times.

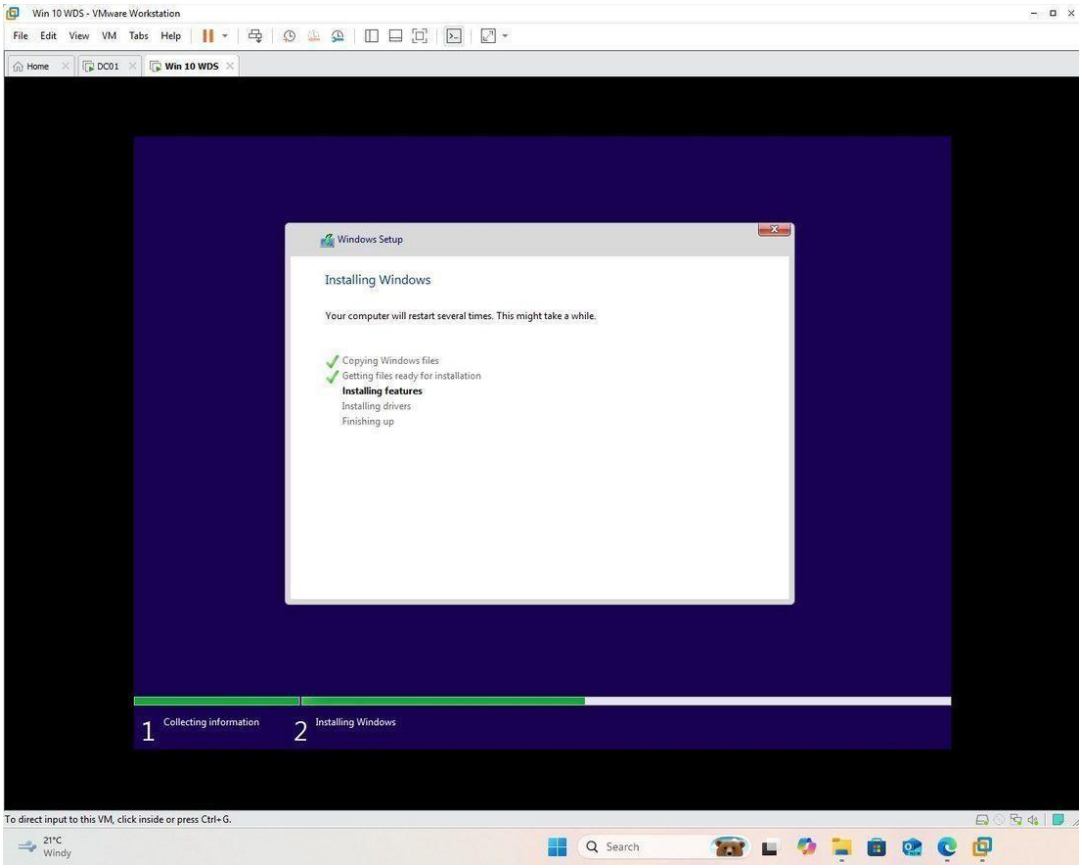


Figure 48: Installing Windows

5.3 First-run setup (OOBE) of the deployed Windows 10

After installation, the deployed computer starts the Windows 10 out-of-box experience, which confirms that the operating system was deployed successfully over the network.

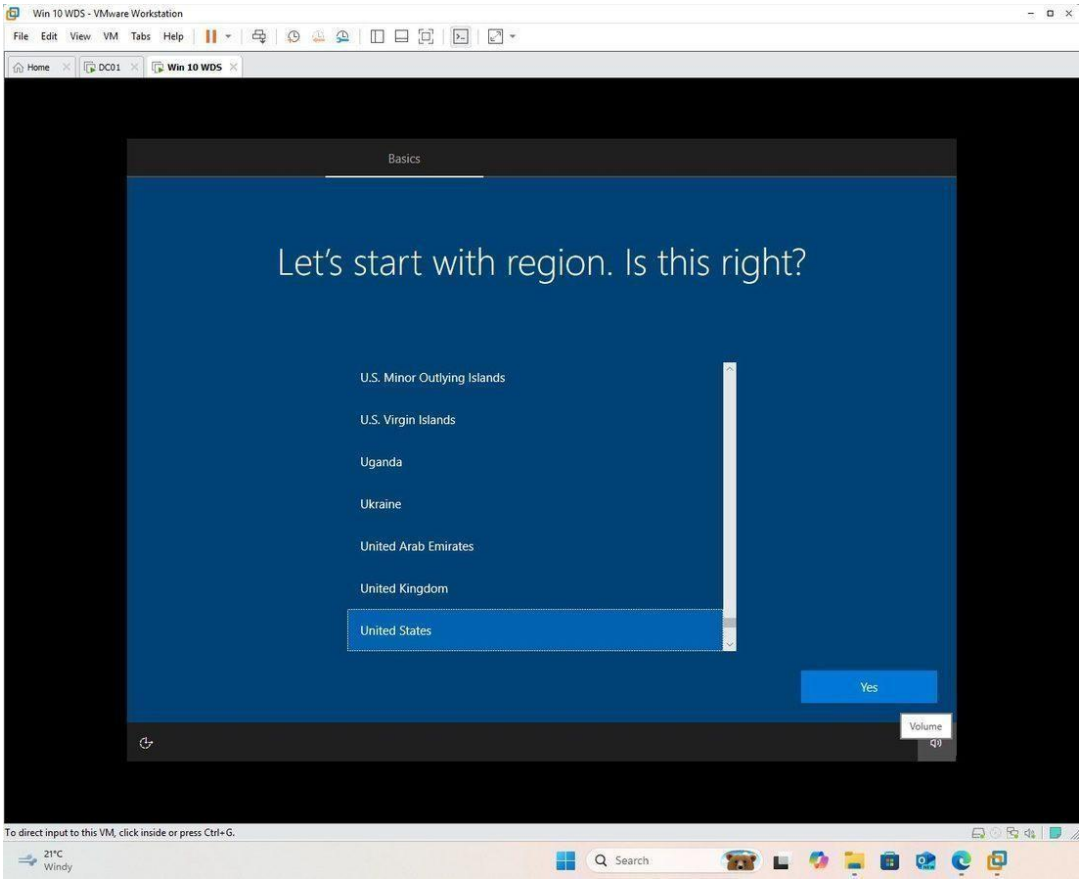


Figure 49: Region selection

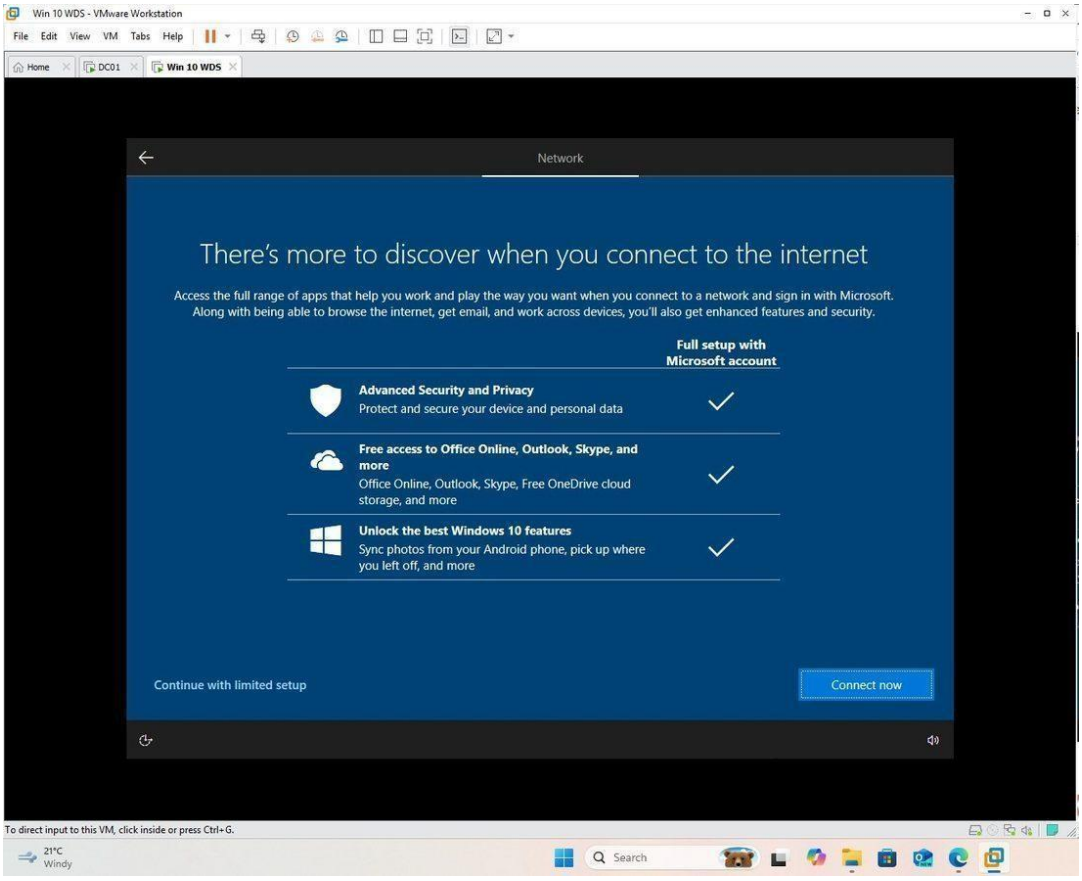


Figure 50: Network page

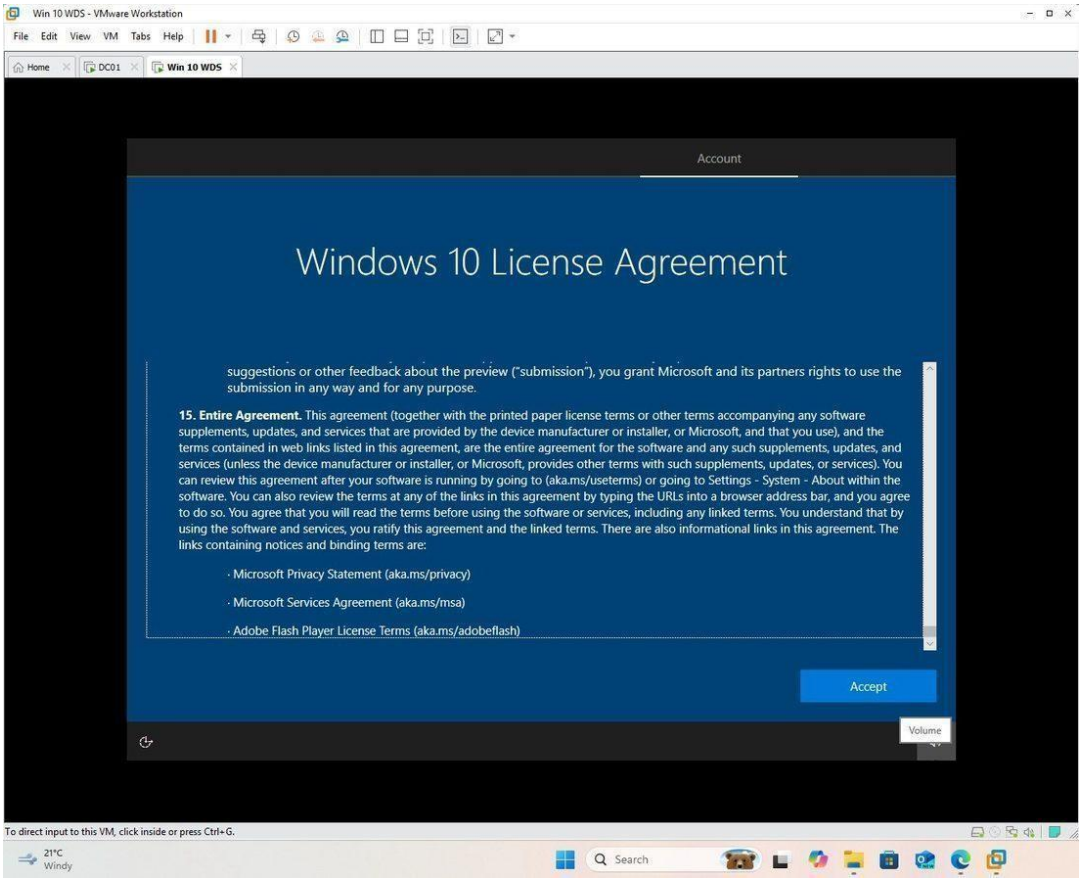


Figure 51: Windows 10 license agreement

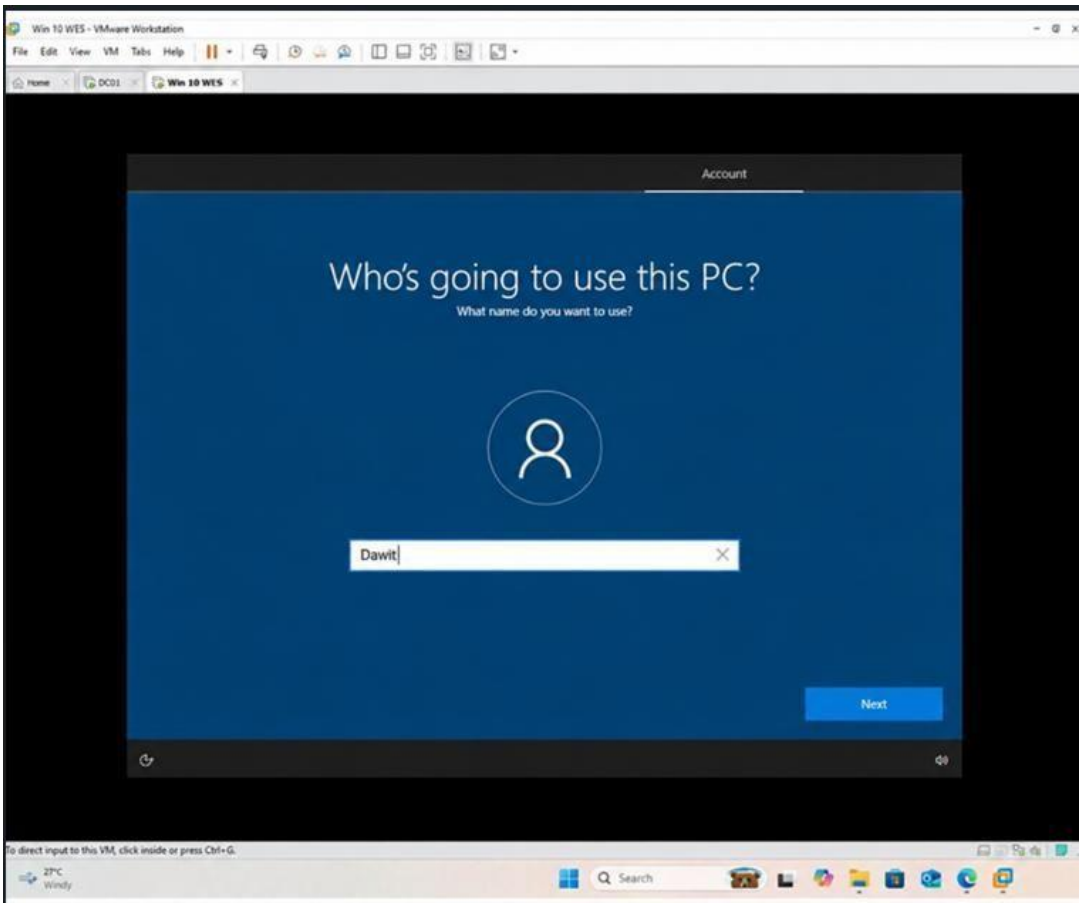


Figure 52: Creating the user account

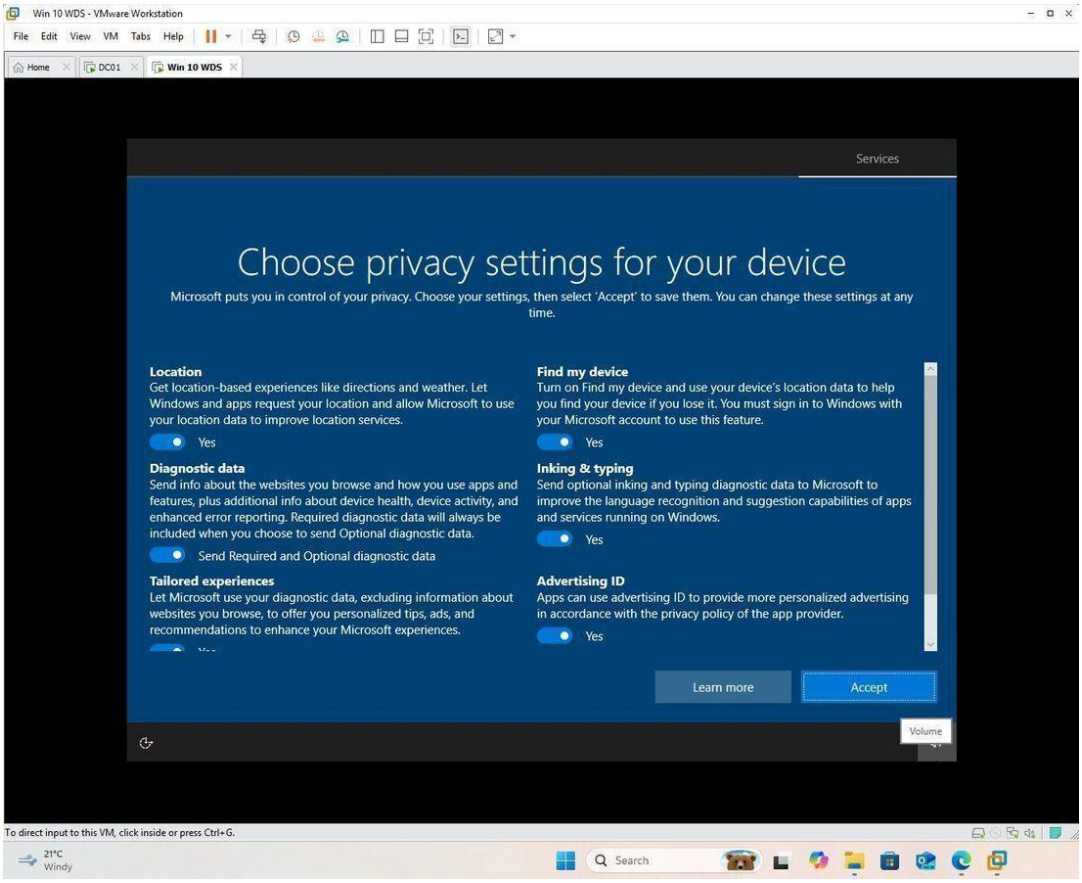


Figure 53: Privacy settings

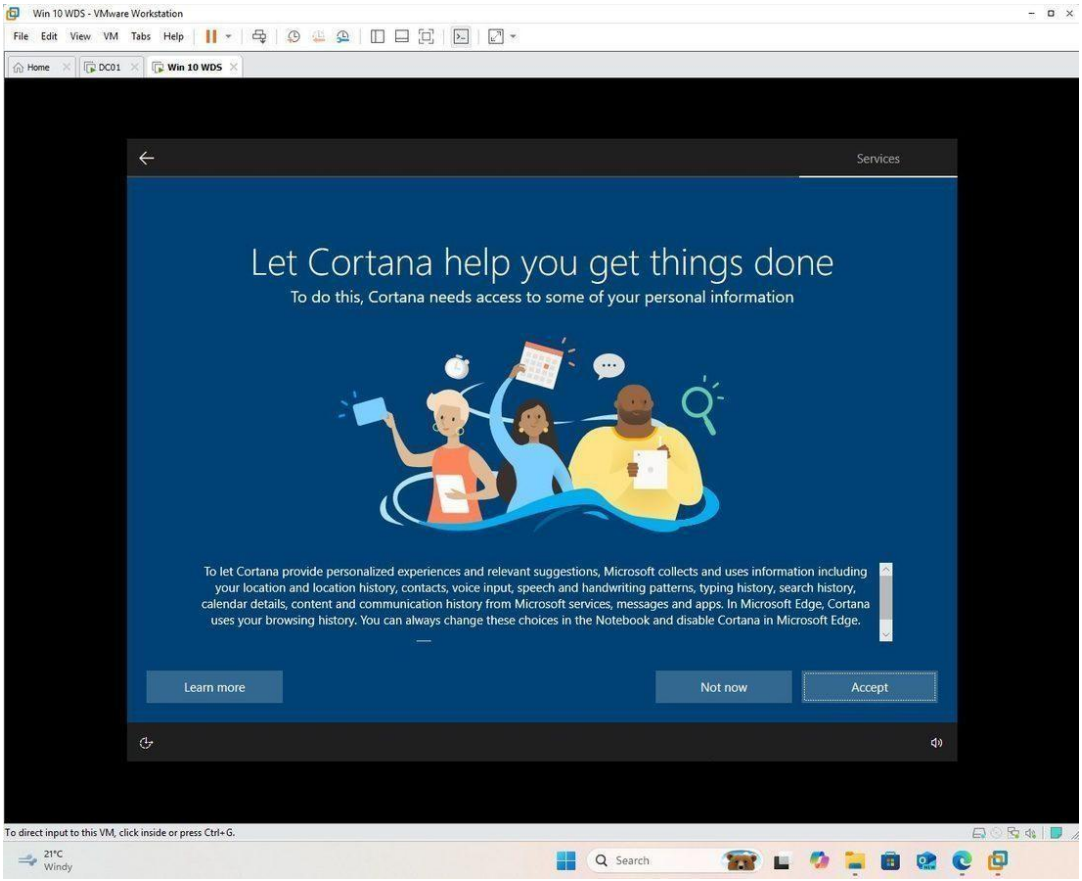


Figure 54: Cortana prompt

6. Results and Verification

- The DHCP Server role was installed and the scope "VLAN 1" (192.168.1.1 to 192.168.1.254, mask 255.255.255.0) was created and activated.
- The exclusion range 192.168.1.1 to 192.168.1.10 and the 1-hour lease duration were configured, and the scope was verified in the DHCP console.
- The WDS role was installed and configured (integrated with Active Directory, responding to all client computers).
- The Windows 10 install image and boot image were added to WDS.
- A blank client VM with no ISO performed a PXE boot, received its configuration from DHCP and WDS, installed Windows 10 over the network, and reached the first-run setup.

7. Conclusion

In this lab, a DHCP server was installed and a scope was configured to assign IP addresses automatically, and Windows Deployment Services was installed to deploy Windows 10 over the network. A blank virtual machine booted through PXE and completed a Windows installation without any USB drive or DVD. This method lets schools and medium-to-large organizations deploy many computers quickly. Custom images can also be created that automatically install drivers and software across the whole network. During the lab, the WDS service timeout was resolved by starting the service manually.

8. Assessment Questions and Answers

1. What is the primary purpose of DHCP in a network environment?

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and related settings, such as subnet mask, default gateway and DNS servers, to client computers. This removes the need to configure each device manually and avoids duplicate or incorrect addresses.

2. Why is it important to authorize a DHCP server in Active Directory Domain Services (ADDS)?

Authorization prevents rogue DHCP servers from handing out wrong addresses and disrupting the network. In an AD domain, a DHCP server that is not authorized will not lease addresses, so the administrator must authorize it (Commit in the post-install wizard) before it can serve clients.

3. Explain the purpose of creating an exclusion range in a DHCP scope.

An exclusion range is a set of addresses inside the scope that the server will not hand out. It is used to reserve addresses for devices with static IPs, such as routers, gateways and servers, so that DHCP never assigns them to clients and causes conflicts. In this lab, 192.168.1.1 to 192.168.1.10 was excluded.

4. What is the significance of setting a lease duration in DHCP? How does it affect network management?

The lease duration defines how long a client may use an address before it must renew it. Short leases (such as 1 hour) suit networks where devices connect and disconnect often, because unused addresses return to the pool quickly, but they increase renewal traffic. Long leases suit stable networks of fixed desktops and reduce traffic, but unused addresses stay reserved longer. The right value helps manage address usage efficiently.

5. What does WDS stand for and what is its primary function?

WDS stands for Windows Deployment Services. It is a Windows Server role that lets administrators deploy Windows operating systems to computers remotely over the network, without needing physical media such as DVDs or USB drives.

6. Why do we need both install.wim and boot.wim files for WDS to function?

The boot.wim file is the boot image: a small Windows PE environment that the client loads over the network to start the setup process. The install.wim file is the install image that contains the actual operating system to be installed. The boot image starts the client and the install image provides the OS, so both are required.

7. What is PXE boot and how does it relate to WDS?

PXE (Pre-Boot Execution Environment) lets a computer boot from its network card instead of a local disk. The client sends a request on the network, receives an IP address and boot server information, and downloads a boot image. WDS acts as the PXE server that provides the boot image and install images to the client.

8. Why is DHCP required for WDS to function?

A PXE client has no operating system or IP address when it starts. It depends on DHCP to receive an IP address and to learn where the boot server is. Without DHCP, the client cannot communicate on the network and cannot reach WDS.

9. In what scenarios would an organization prefer using WDS over traditional USB installation methods?

WDS is preferred when many computers must be deployed quickly, such as in schools, computer labs, and medium-to-large businesses. It is also useful when machines are in different locations, when a standard configuration is needed on all machines, and when reinstalling or reimaging computers regularly. It saves time and removes the need to carry physical media to each computer.

10. What are the advantages of creating custom images for deployment through WDS?

Custom images contain the operating system together with preconfigured settings, drivers and software, so every computer receives the same standard configuration automatically. This saves installation time, reduces errors and manual work, improves consistency and security, and makes it easy to deploy across the entire network.